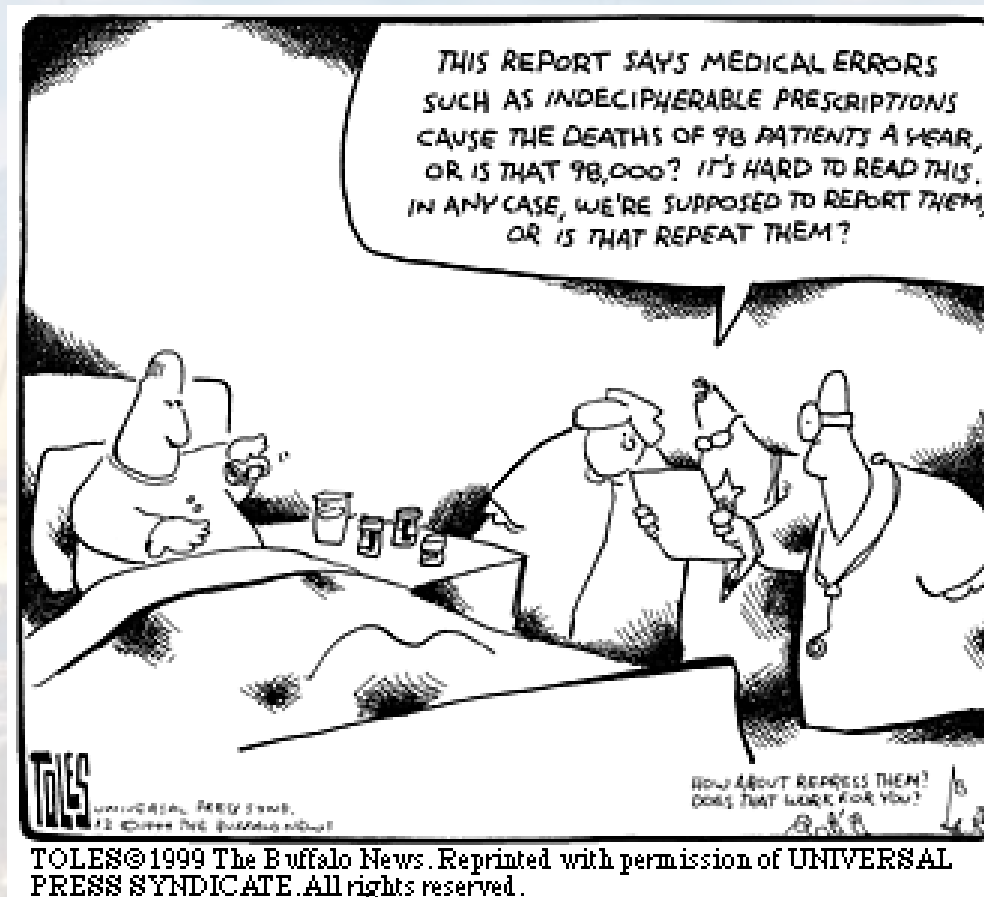


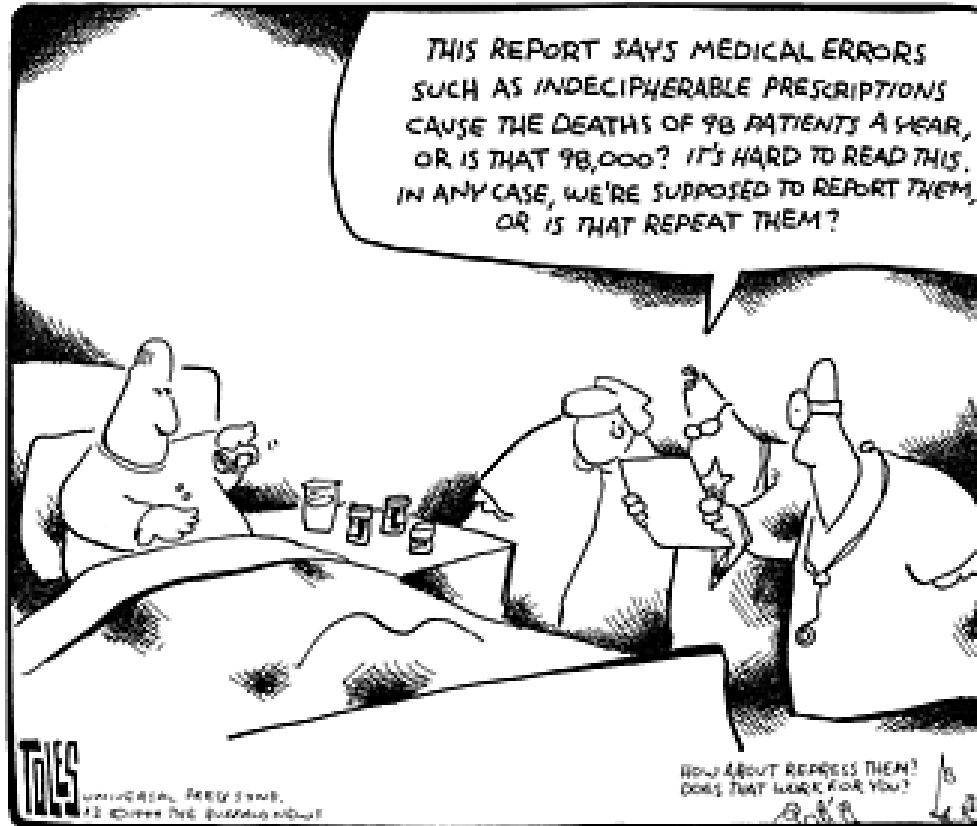
M. Hohle:

Physics 88: Introduction to Computational Techniques in Physics





Week	Date	Topic
1	Aug 26 th	Programming Environment & UIs for Python. Programming Fundamentals
2	Sep 2 nd	Basic Types in Python
3	Sep 9 th	Parsing, Data Processing and File I/O/ Visualization: Matplotlib & Seaborn
4	Sep 16 th	Functions, Map & Lambda
5	Sep 23 rd	Scientific Computing with numpy and avoiding Loops
6	Sep 30th	Random Numbers & Probability Distributions
7	Oct 7 th	Interpreting Measurements & Introduction to Bayesian Statistics
8	Oct 14 th	Numerical Integration and Differentiation
9	Oct 21 st	Root finding, Interpolation
10	Oct 28 th	Systems of Linear Equations
11	Nov 4 th	Introduction to Ordinary Differential Equations (ODEs)
	Nov 11 th	<i>Veteran's Day</i>
12	Nov 18 th	Stability of ODEs, Runge-Kutta, Examples
	Nov 25 th	<i>Thanksgiving</i>
13	Dec 2 nd	Final Project Presentations



Outline:

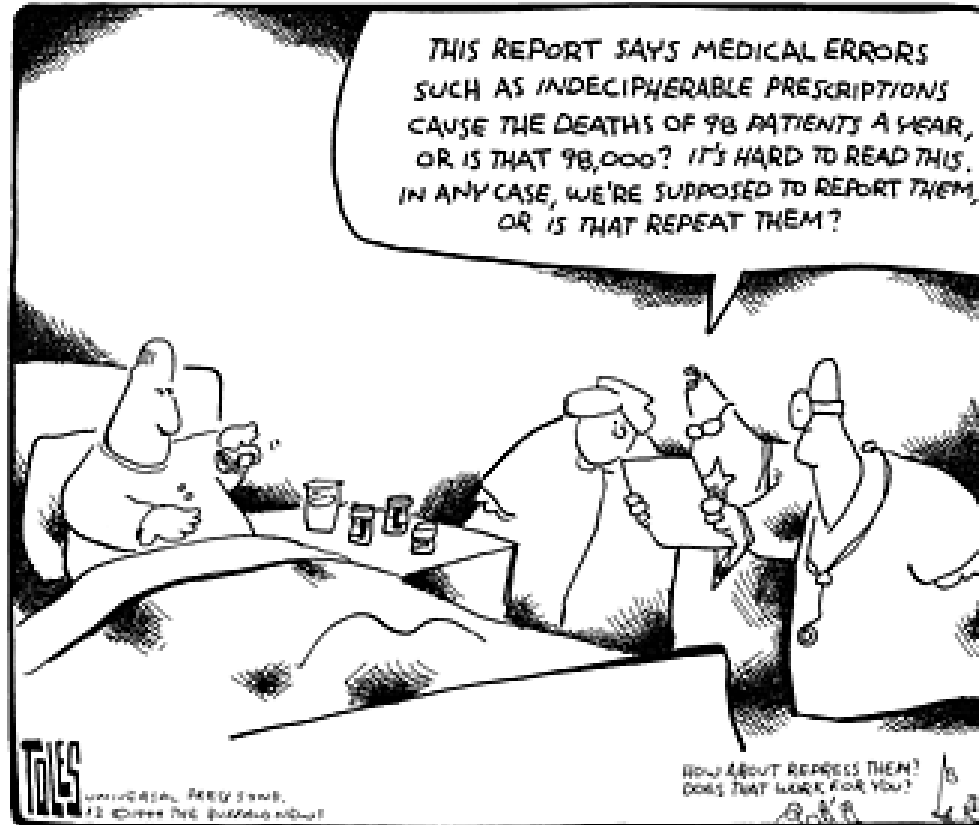
Basics

Most Common PDFs

- uniform
- binomial
- Poissonian
- Normal/Gaussian

Error Estimation

Bayesian Statistics



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Outline:

Basics

Most Common PDFs

- uniform
- binomial
- Poissonian
- Normal/Gaussian

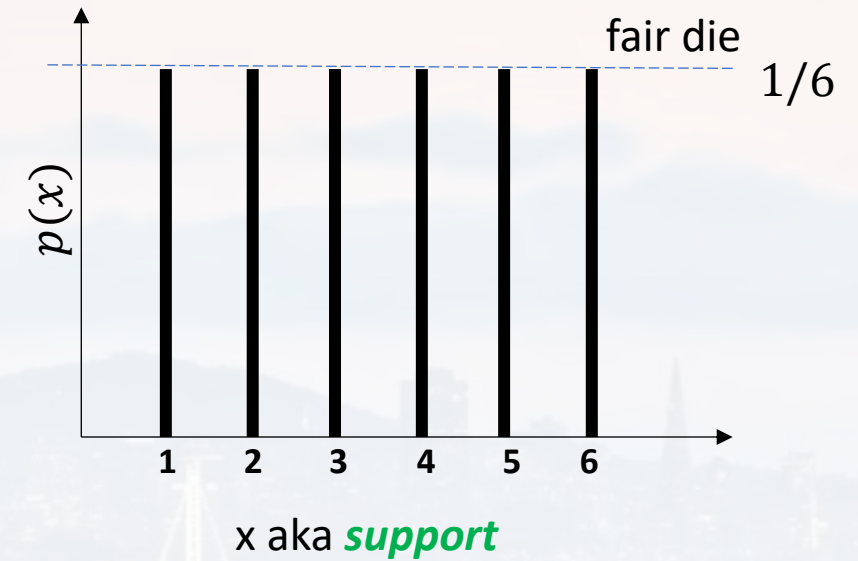
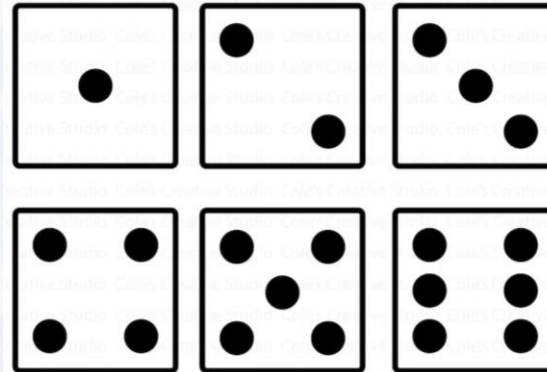
Error Estimation

Bayesian Statistics



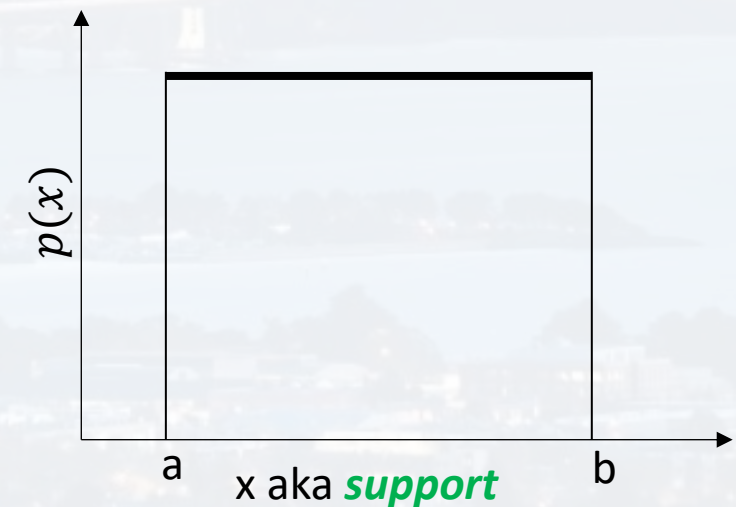
distributions

discrete (= countable)



continuous

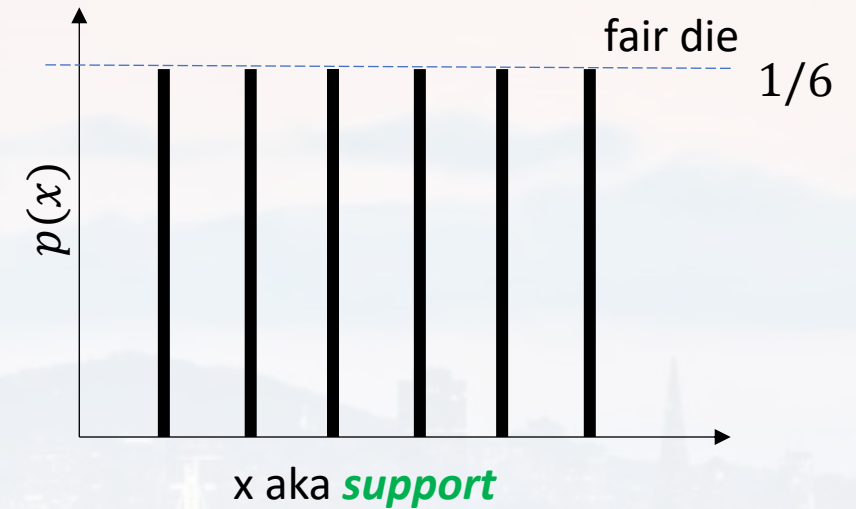
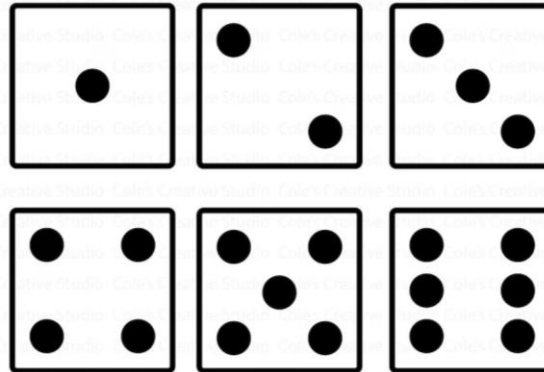
$$[a \leq x \leq b]$$





distributions

discrete (= countable)



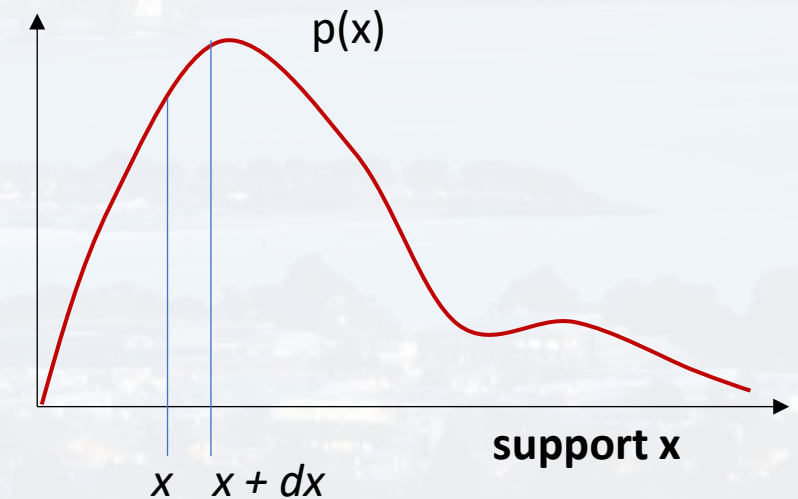
continuous

$$[a \leq x \leq b]$$

$p(x)$ doesn't make sense

$$\rightarrow p(x) dx$$

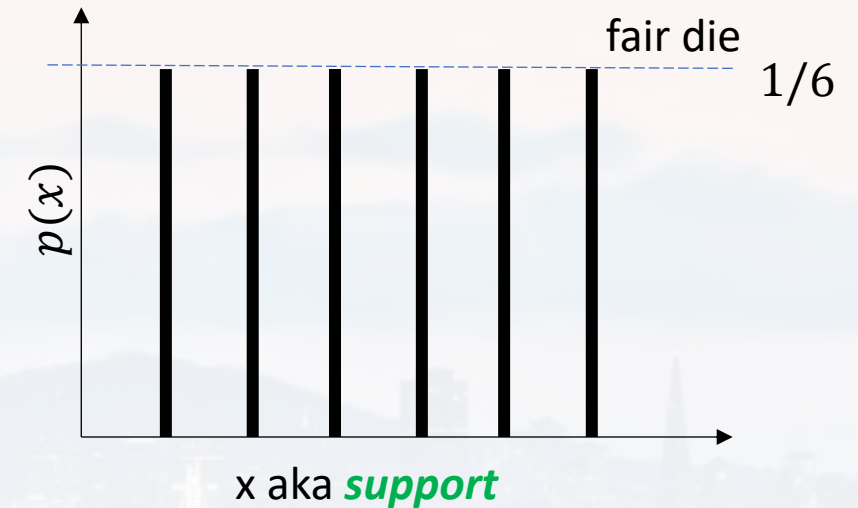
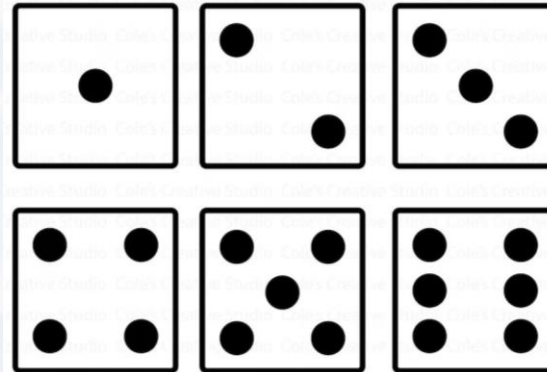
probability **d**ensity **f**unction





distributions

discrete (= countable)



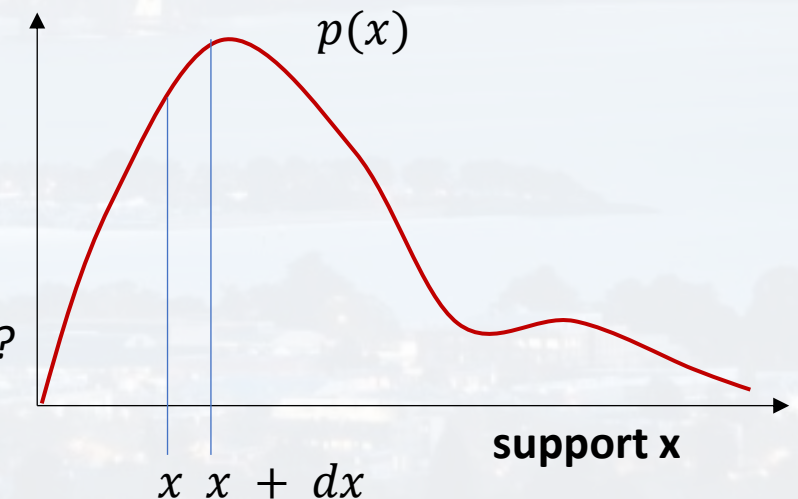
continuous

$p(x)$ doesn't make sense
 $\rightarrow p(x) dx$

probability **d**ensity **f**unction

dx defines the probability!

*What is the probability to find a person who is **EXACTLY** 6ft tall?
Depends on how **accurate** (dx) you measure!*





the mean μ

(barycenter)

the variance σ^2

(natural scatter)

discrete (= countable)

$$\mu = E(x) = \sum_i x_i p(x_i)$$

$$\sigma^2 = \text{var}(x) = \sum_i (x_i - \mu)^2 p(x_i)$$

continuous

$$\mu = E(x) = \int x p(x) dx$$

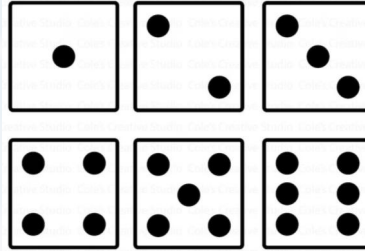
$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$



uniform distribution: discrete vs continuous

the mean μ

discrete (= countable)



$$\mu = \sum_i x_i p(x_i) = \sum_{i=1}^6 i \frac{1}{6} = \mathbf{3.5}$$

continuous

$$[a \leq x \leq b]$$

$$\mu = \int x p(x) dx$$

$p(x) = \text{const}$
(uniform)

$$= \text{const} \int_a^b x dx = \text{const} \frac{1}{2} (b^2 - a^2)$$

2nd axiom

$$\int_a^b p(x) dx = 1$$

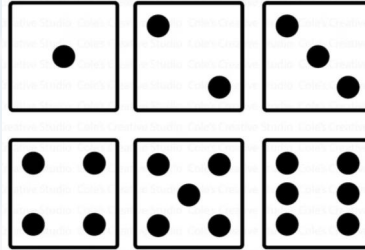
$$\text{const} \int_a^b dx = 1 \quad \rightarrow \text{const} = \frac{1}{b - a}$$



uniform distribution: discrete vs continuous

the mean μ

discrete (= countable)



$$\mu = \sum_i x_i p(x_i) = \sum_{i=1}^6 i \frac{1}{6} = \mathbf{3.5}$$

continuous

$$[a \leq x \leq b]$$

$$\mu = \int x p(x) dx = \text{const} \frac{1}{2} (b^2 - a^2)$$

$$\text{const} = \frac{1}{b - a}$$

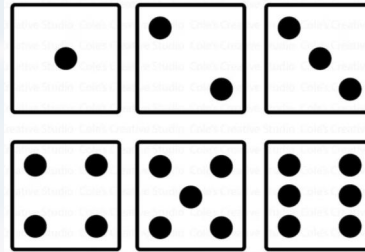
$$\mu = \frac{1}{2} \frac{b^2 - a^2}{b - a}$$

= **3.5** for $a = 1$ and $b = 6$



the variance σ^2

discrete (= countable)



$$\sigma^2 = \sum_i (x_i - \mu)^2 p(x_i) = \frac{1}{6} \sum_{i=1}^6 (i - 3.5)^2 \approx \mathbf{2.9}$$

continuous

$[a \leq x \leq b]$

$$\sigma^2 = \int (x - \mu)^2 p(x) dx = \frac{1}{b-a} \int_a^b (x - 3.5)^2 dx$$

$$\sigma^2 = \frac{1}{12} (b - a)^2$$

$$= 25/12 \approx \mathbf{2.1} \text{ for } a = 1 \text{ and } b = 6$$



$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

$$\text{var}(x) = \int (x - \mu)^2 p(x) dx = E([x - \mu]^2)$$

variance can be interpreted as **mean of $[x - \mu]^2$**

$$= E(x^2 - 2x\mu + \mu^2)$$

$$= \int [x^2 - 2x\mu + \mu^2] p(x) dx$$

$$= \int x^2 p(x) dx - 2\mu \int x p(x) dx + \mu^2 \int p(x) dx$$

$$= E(x^2) - 2\mu E(x) + \mu^2 E(1)$$

2nd axiom $\int p(x) dx = 1$

$$= E(x^2) - 2\mu E(x) + \mu^2$$

$$\mu = E(x)$$

$$\sigma^2 = E(x^2) - E(x)^2$$



$$\sigma^2 = E(x^2) - E(x)^2$$

plotting two sets of random number: x_1 and x_2

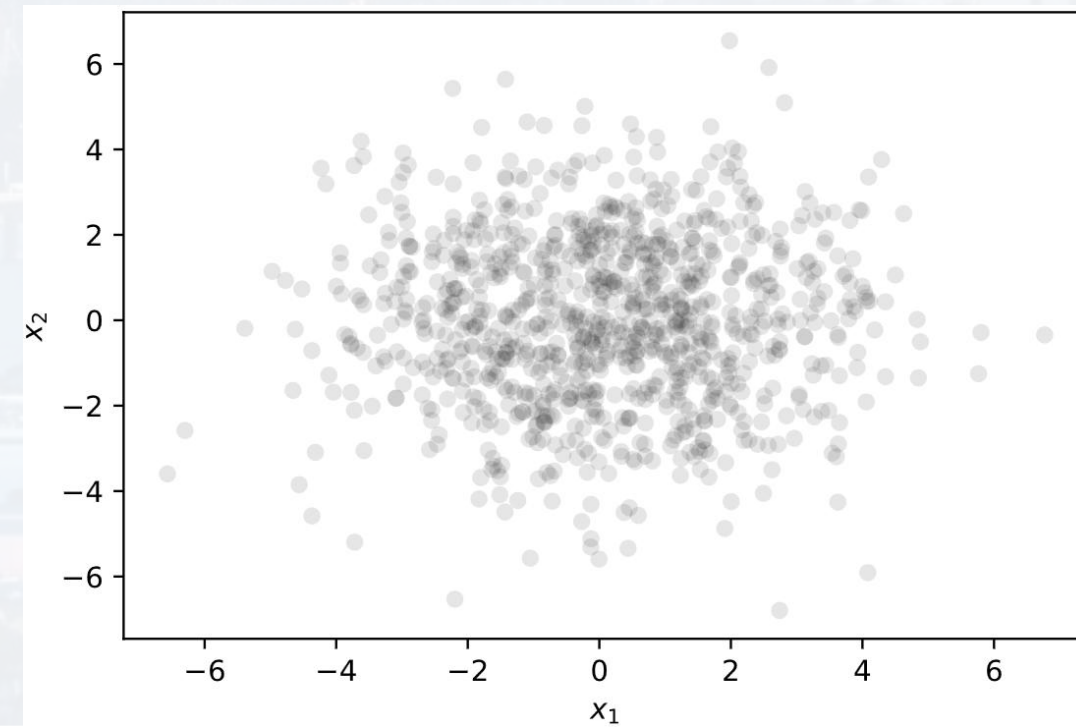
```
x1 = np.random.normal(0,2,(1000,))  
x2 = np.random.normal(0,2,(1000,))
```

```
plt.scatter(x1, x2, color = 'k', alpha = 0.1, edgecolor = 'none')  
plt.xlabel('$x_1$')  
plt.ylabel('$x_2$')
```

x_1 and x_2 are unrelated and mutually **independent**
→ featureless data cloud

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = var(x) = \int (x - \mu)^2 p(x) dx$$





$$\sigma^2 = E(x^2) - E(x)^2$$

plotting two sets of random number: x_1 and x_2

```
x1 = np.random.normal(0,2,(1000,))
```

```
x2 = np.random.normal(0,20,(1000,))
```

```
plt.scatter(x1, x2, color = 'k', alpha = 0.1, edgecolor = 'none')
```

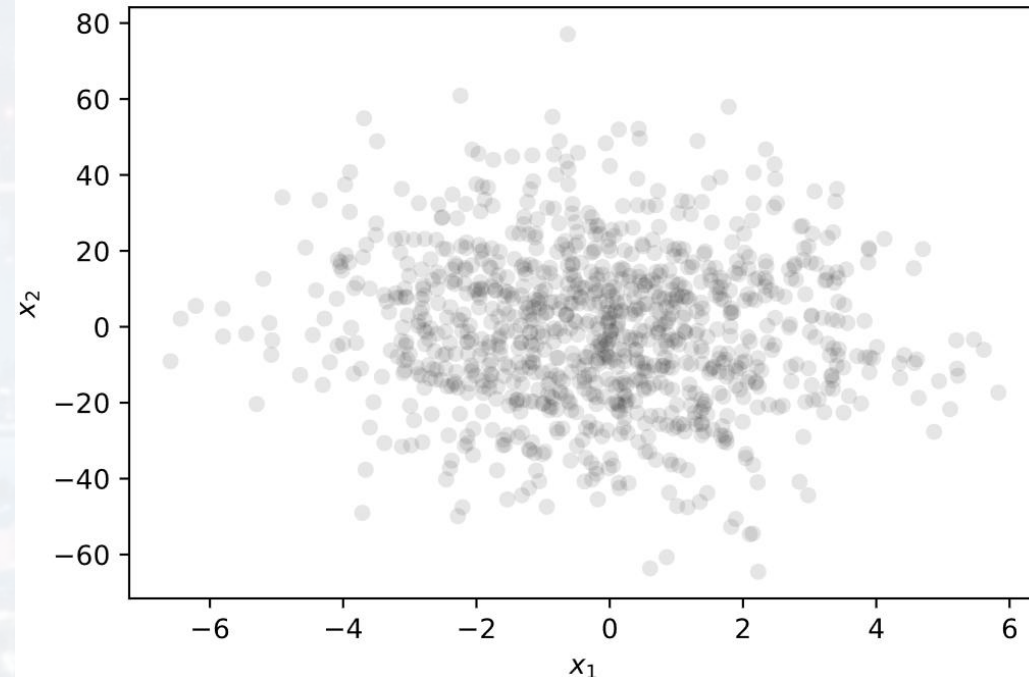
```
plt.xlabel('$x_1$')
```

```
plt.ylabel('$x_2$')
```

x_1 and x_2 are unrelated and mutually **independent**
→ featureless data cloud

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = var(x) = \int (x - \mu)^2 p(x) dx$$





$$\sigma^2 = E(x^2) - E(x)^2$$

plotting two sets of random number: x_1 and x_2

```
x1 = np.random.normal(0,2,(1000,))  
x2 = x1**2 + x1  
#x2 = 4*x1
```

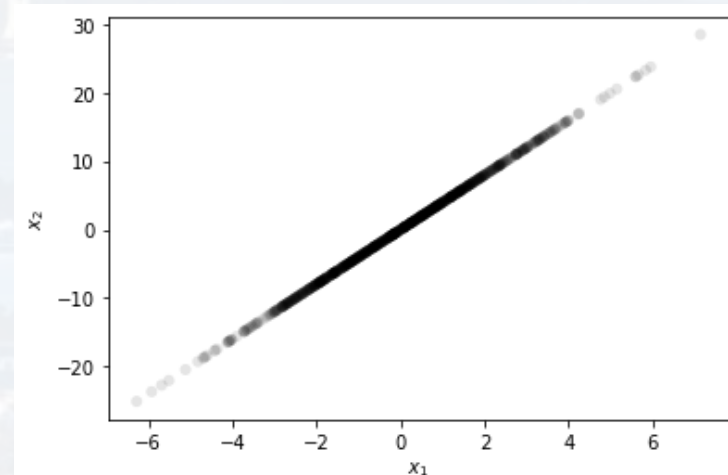
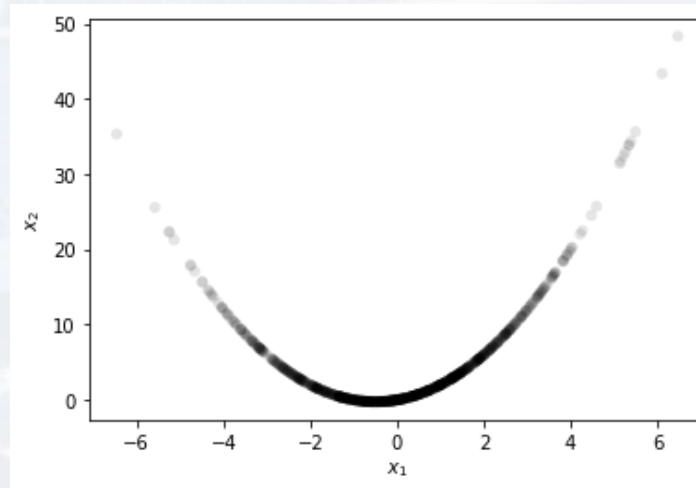
```
plt.scatter(x1, x2, color = 'k', alpha = 0.1, edgecolor = 'none')  
plt.xlabel('$x_1$')  
plt.ylabel('$x_2$')
```

based on the shape of the data cloud

→ prediction how x_1 and x_2 are related, i. e. how they **correlate**

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = var(x) = \int (x - \mu)^2 p(x) dx$$





$$\sigma^2 = E(x^2) - E(x)^2$$

```
x1 = np.random.normal(0,2,(1000,))
```

```
x2 = np.random.normal(3,3,(1000,))
```

```
x3 = np.random.uniform(0,5,(1000,))
```

```
x4 = 5*np.random.uniform(3,4,(1000,))
```

```
x5 = np.sqrt(x4)
```

```
x6 = x1 + x2
```

```
x7 = 2*x3
```

```
x8 = x3*x2
```

```
All = np.vstack((x1, x2, x3, x4, x5, x6, x7, x8))
```

```
data = pd.DataFrame(All.transpose(),
```

```
                      columns = ['x1', 'x2', 'x3', 'x4', 'x5', 'x6', 'x7', 'x8'])
```

```
out = sns.pairplot(data, kind = "kde", \
```

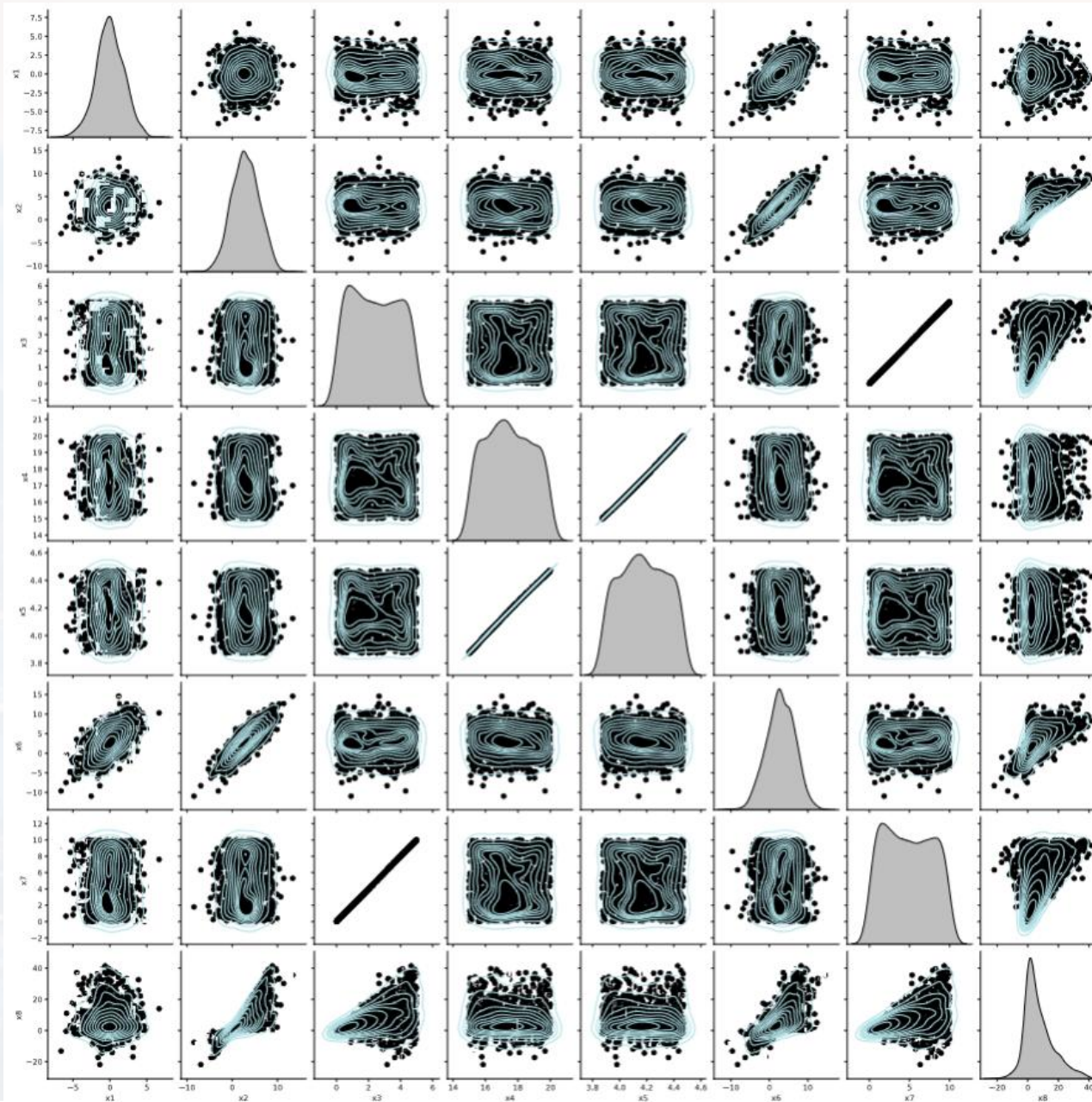
```
                      plot_kws = {'color':[176/255, 224/255, 230/255]}, \
```

```
                      diag_kws = {'color':'black'})
```

```
out.map_offdiag(plt.scatter, color = 'black')
```

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = var(x) = \int (x - \mu)^2 p(x) dx$$



$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

based on the shape of the data cloud

→ prediction how x_1 and x_2 are related, i. e.
how they **correlate**

→ how to quantify?



$a, b = \text{const}$

$$\sigma^2 = E(x^2) - E(x)^2$$

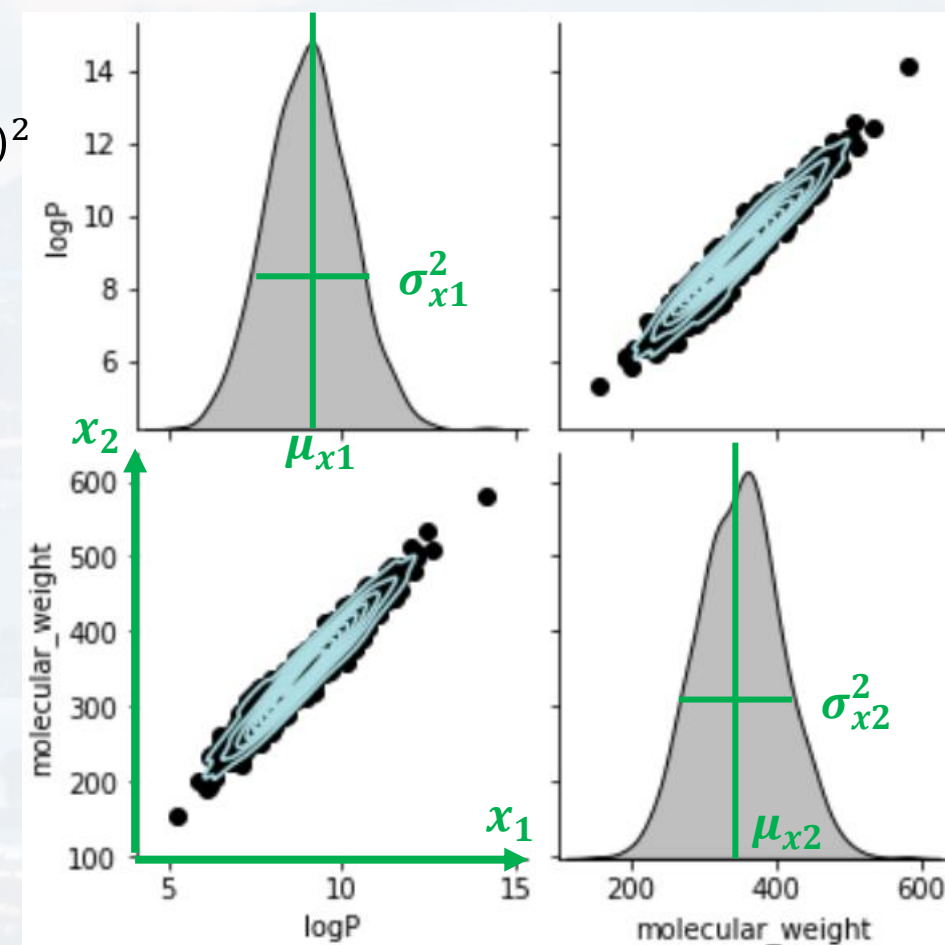
$$\text{var}([a x_1 + b x_2]) = E([a x_1 + b x_2]^2) - E(a x_1 + b x_2)^2$$

$$= E(a^2 x_1^2 + 2ab x_1 x_2 + b^2 x_2^2) - E(a x_1 + b x_2)^2$$

$$= a^2 E(x_1^2) + 2ab E(x_1 x_2) + b^2 E(x_2^2) - E(a x_1 + b x_2)^2$$

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$





$$a, b = \text{const}$$

$$\sigma^2 = E(x^2) - E(x)^2$$

$$\mu = E(x) = \int x p(x) dx$$

$$\text{var}([a x_1 + b x_2]) = E([a x_1 + b x_2]^2) - E(a x_1 + b x_2)^2$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

$$= E(a^2 x_1^2 + 2ab x_1 x_2 + b^2 x_2^2) - E(a x_1 + b x_2)^2$$

$$= a^2 E(x_1^2) + 2ab E(x_1 x_2) + b^2 E(x_2^2) - E(a x_1 + b x_2)^2$$

$$= a^2 E(x_1^2) + 2ab E(x_1 x_2) + b^2 E(x_2^2) - [aE(x_1) + bE(x_2)]^2$$

$$= a^2 E(x_1^2) - a^2 E(x_1)^2 + b^2 E(x_2^2) - b^2 E(x_2)^2 + 2ab E(x_1 x_2) - 2ab E(x_1)E(x_2)$$

$$a^2 \text{var}(x_1)$$

$$b^2 \text{var}(x_2)$$

$$2ab \text{cov}(x_1, x_2)$$

$$= a^2 \text{var}(x_1) + b^2 \text{var}(x_2) + 2ab \text{cov}(x_1, x_2)$$

$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

covariance



$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

What does the covariance tell us?

- 1) geometrical interpretation → next lectures
- 2) arithmetical interpretation

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$



$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

What does the covariance tell us?

1) geometrical interpretation → next lectures

2) arithmetical interpretation

a) x_1 and x_2 are independent

$$E(x_1 x_2) - E(x_1)E(x_2)$$

$$= \iint x_1 x_2 p(x_1) p(x_2) dx_1 dx_2 - \int x_1 p(x_1) dx_1 \int x_2 p(x_2) dx_2$$

x_1 and x_2 are independent:

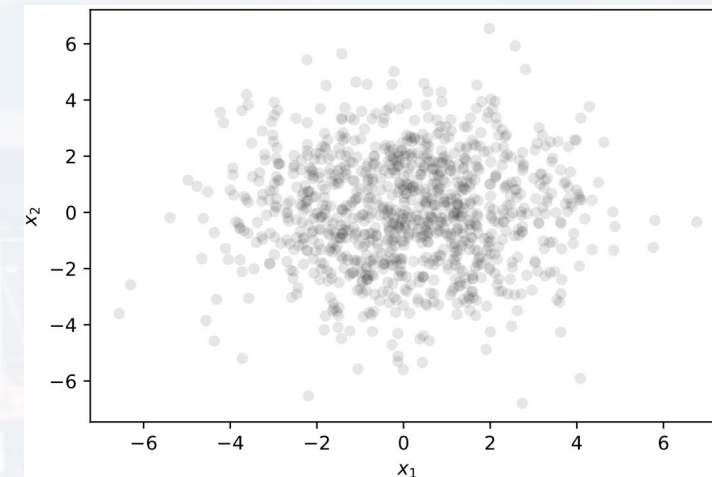
x_1 is not a function of x_2 and vice versa

x_1 cannot be predicted by x_2 and vice versa

$$= \int x_1 p(x_1) dx_1 \int x_2 p(x_2) dx_2 - \int x_1 p(x_1) dx_1 \int x_2 p(x_2) dx_2 = 0$$

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$



Covariance equals **zero**
if samples are **independent**!



$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

What does the covariance tell us?

1) geometrical interpretation → next lectures

2) arithmetical interpretation

b) x_1 and x_2 are **not** independent

$$E(x_1 x_2) - E(x_1)E(x_2)$$

$$= \iint x_1 x_2 p(x_1) p(x_2) dx_1 dx_2 - \int x_1 p(x_1) dx_1 \int x_2 p(x_2) dx_2$$

x_1 and x_2 are **not** independent:

x_1 **is** a function of x_2 and vice versa

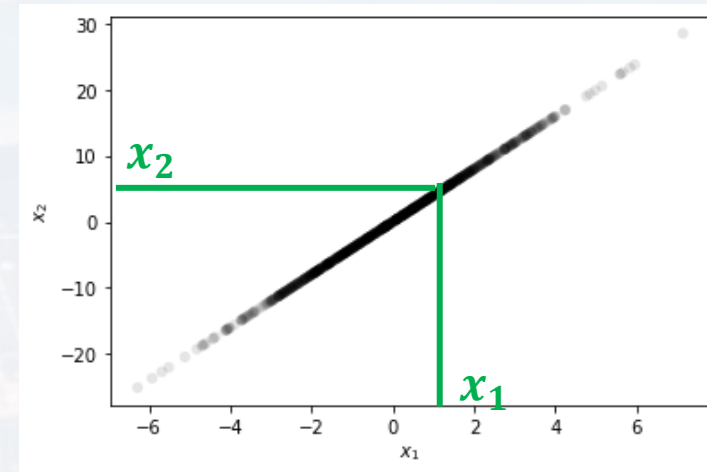
x_1 **can** be predicted by x_2 to certain degree and vice versa

$$= \iint x_1 p(x_1) x_2(x_1) p(x_2(x_1)) dx_1 dx_2(x_1) - \int x_1 p(x_1) dx_1 \int x_2 p(x_2) dx_2$$

Covariance **does not**
equal **zero**!

$$\mu = E(x) = \int x p(x) dx$$

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

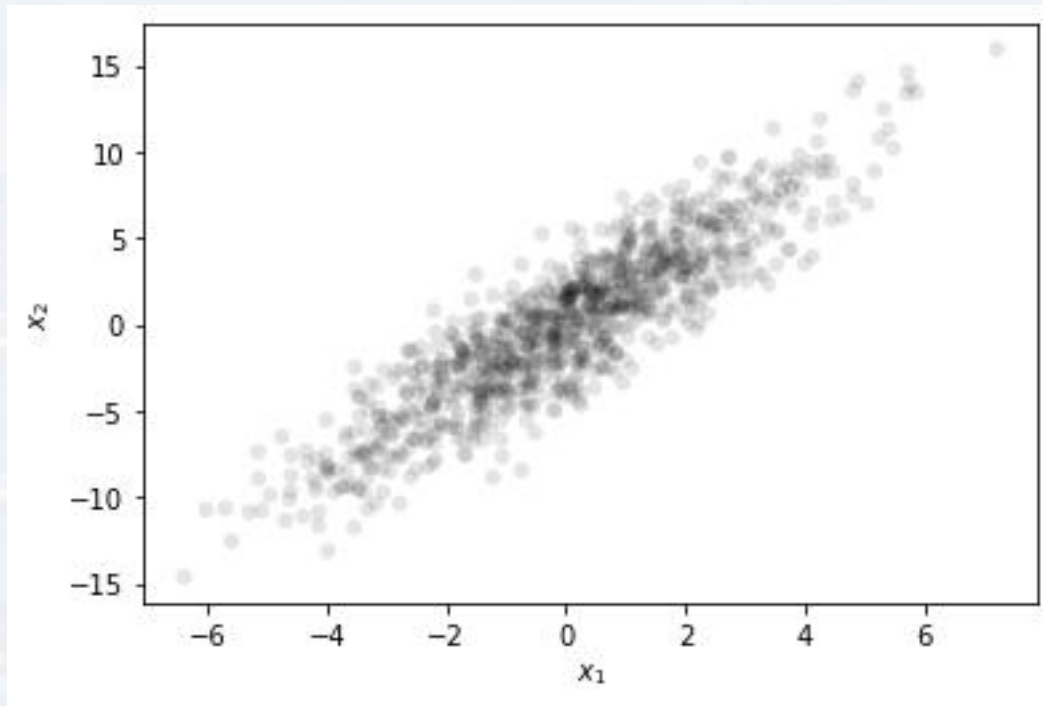




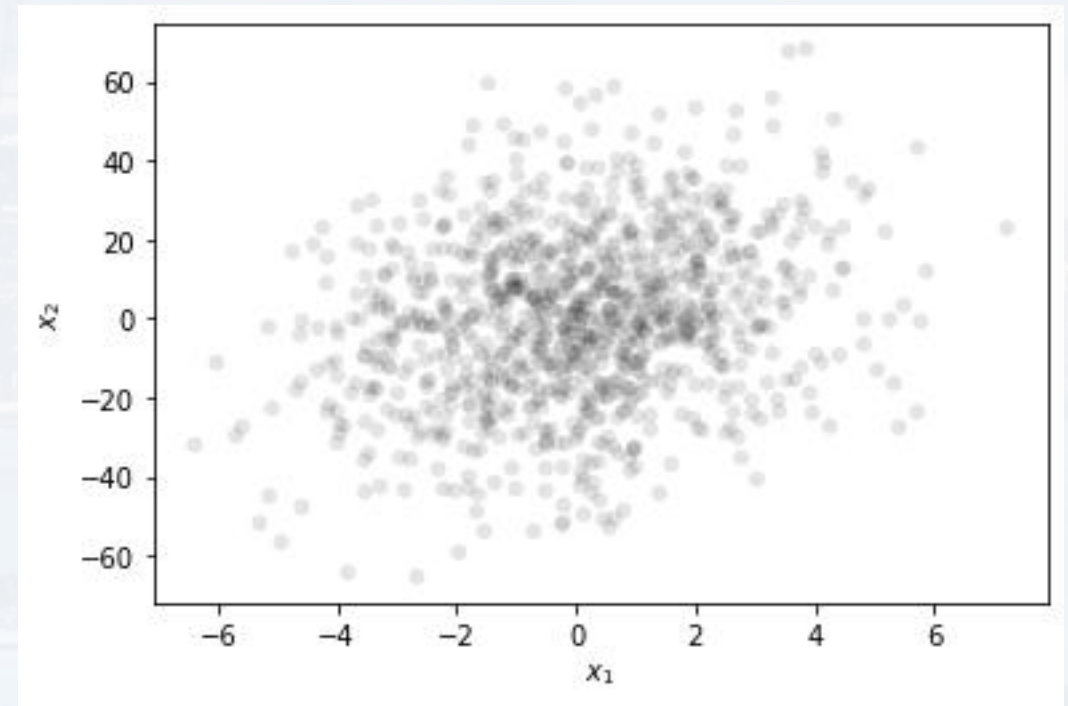
$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

covariance

```
x1 = np.random.normal(0, 2, (1000,))  
x2 = 2*x1 + np.random.normal(0, 2, (1000,))
```



```
x1 = np.random.normal(0, 2, (1000,))  
x2 = 2*x1 + np.random.normal(0, 20, (1000,))
```

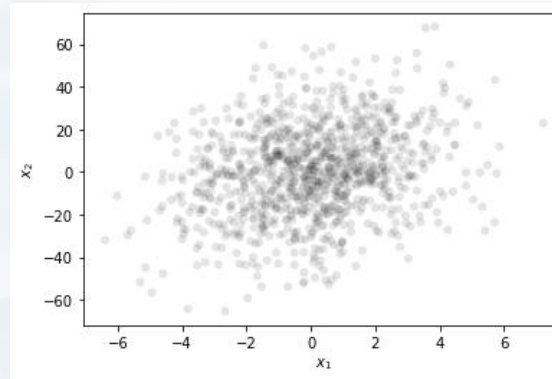
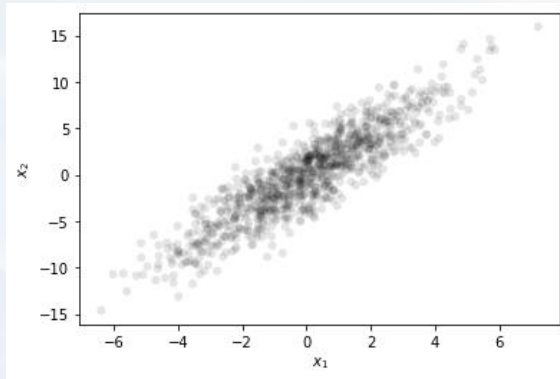


Same dependency, but different variance!



$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

covariance



Same dependency, but different variance!

Need to scale for the variance!

Pearson's correlation coefficient

$$\rho(x_1, x_2) = \frac{\text{cov}(x_1, x_2)}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

$\rho(x_1, x_2)$:

- ranges from -1 to +1
- zero: no correlation
(completely independent)
- -1: max anti correlation
- +1: max correlation



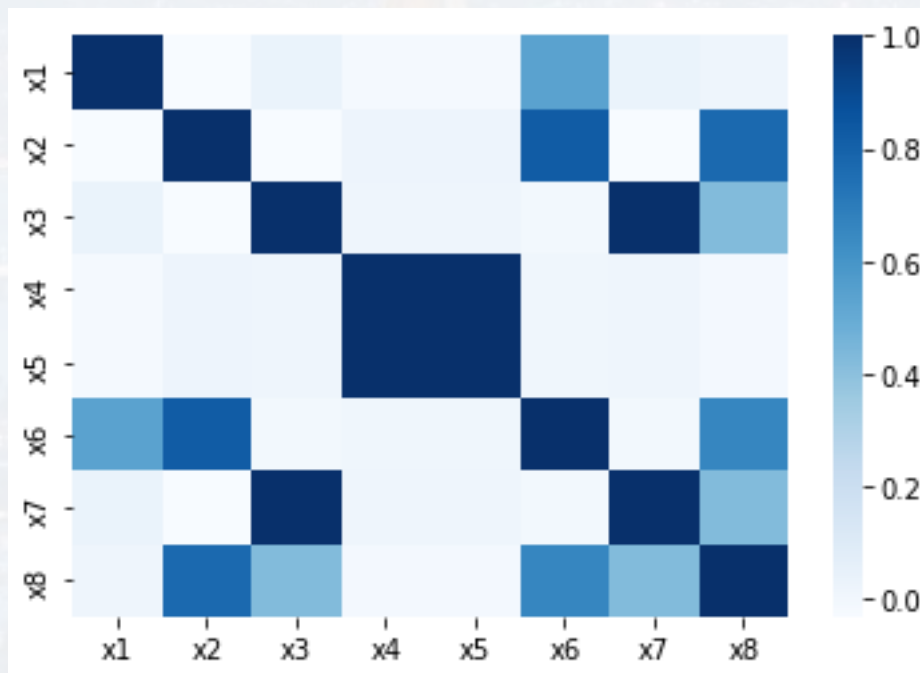
$$\text{cov}(x_1, x_2) = \text{cov}(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

covariance

$$\rho(x_1, x_2) = \frac{\text{cov}(x_1, x_2)}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

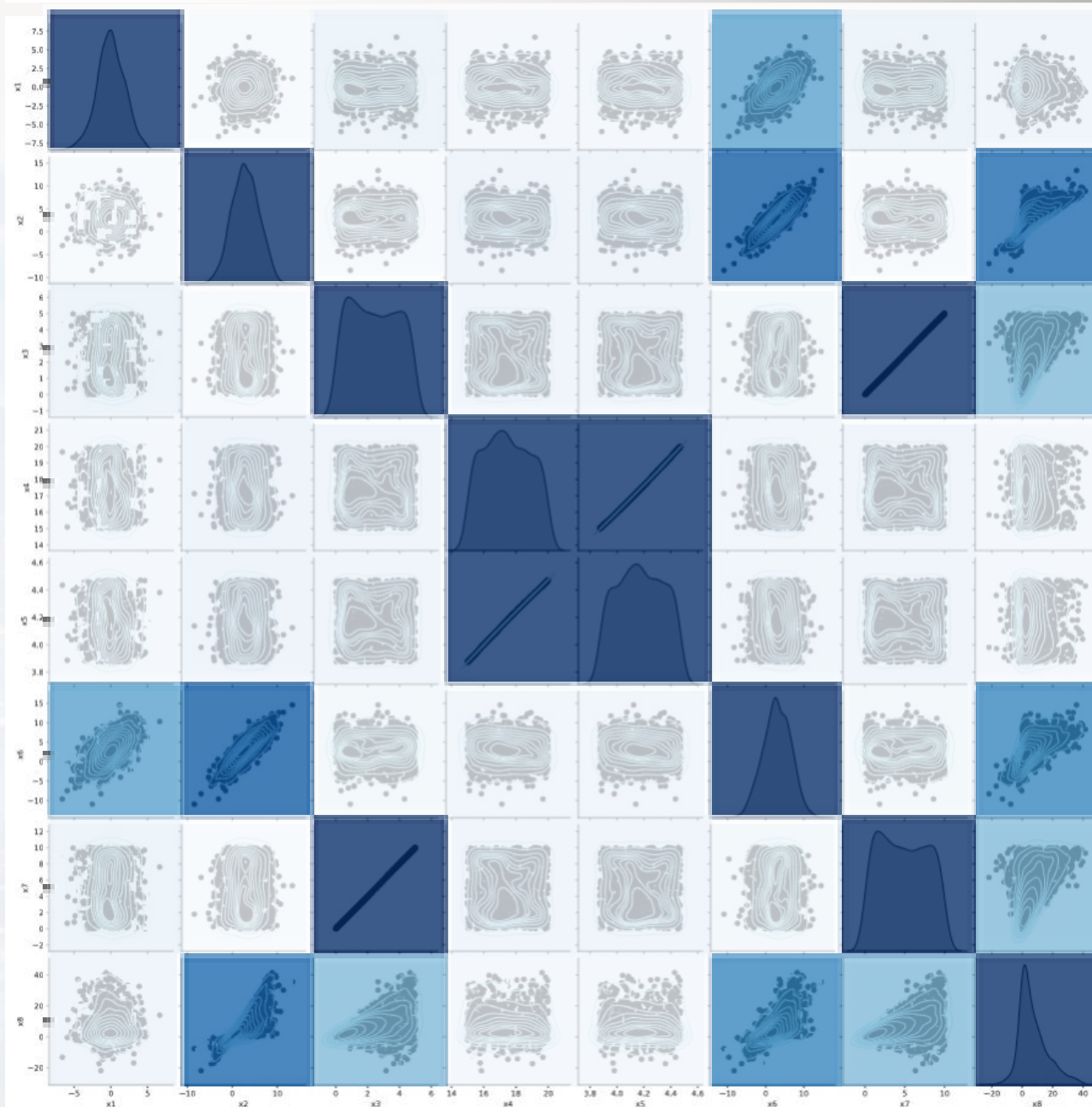
Pearson's correlation
coefficient

```
sns.heatmap(data.corr(), cmap = "Blues")
```



$\rho(x_1, x_2)$:

- ranges from -1 to +1
- zero: no correlation
(completely independent)
- -1: max anti correlation
- +1: max correlation



$\rho(x_1, x_2)$:

- ranges from -1 to +1
- zero: no correlation
(completely independent)
- -1: max anti correlation
- +1: max correlation



Important quantities you should know:

mean

$$\mu = E(x) = \int x p(x) dx$$

median m

$$\int_a^m p(x) dx = \frac{1}{2}$$

variance

$$\sigma^2 = \text{var}(x) = \int (x - \mu)^2 p(x) dx$$

$$\sigma^2 = E(x^2) - E(x)^2$$

$$\sigma_{tot}^2 = \sigma_1^2 + \sigma_2^2 + 2 \text{cov}(x_1, x_2)$$

covariance

$$\text{cov}(x_1, x_2) = E(x_1 x_2) - E(x_1)E(x_2)$$

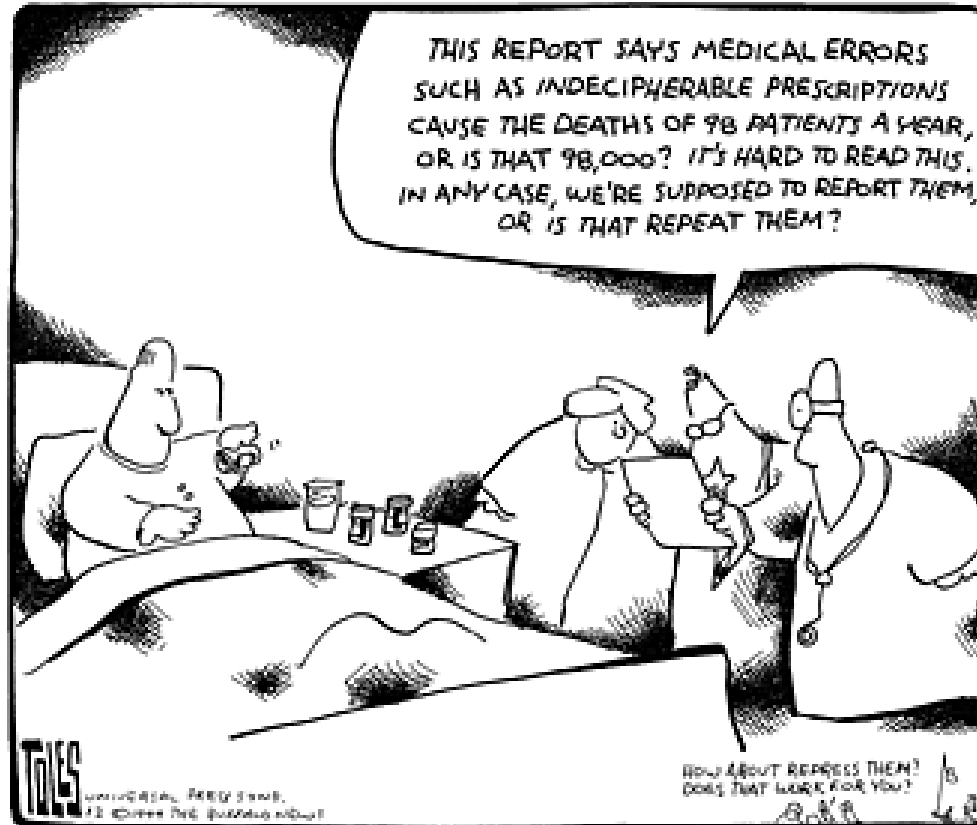
**correlation
coefficient**

$$\rho(x_1, x_2) = \frac{\text{cov}(x_1, x_2)}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

note:

$$\int (x - \mu)^n p(x) dx$$

called n -th *moment*
of a pdf



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Outline:

Basics

Most Common PDFs

- uniform
- binomial
- Poissonian
- Normal/Gaussian

Error Estimation

Bayesian Statistics



finding those $p(x)$ that maximize the **entropy** S , given constraints $\mathbf{C}$

$$S = - \int p(x) \ln[p(x)] dx \quad \mathbf{c:} \quad \int p(x) dx = 1 \quad \longrightarrow \quad p(x) = \text{const}$$

$$\mu = \int_a^b x p(x) dx = \text{const} \int_a^b x dx = \text{const} \frac{1}{2} (b^2 - a^2)$$

$$\int_a^b p(x) dx = 1 \quad \text{const} \int_a^b dx = 1 \quad \rightarrow \text{const} = \frac{1}{b - a}$$

$$\mu = \frac{1}{2} \frac{b^2 - a^2}{b - a}$$

$$\sigma^2 = \frac{1}{12} (b - a)^2$$

Note: the uniform distribution has the largest entropy

→ maximum ignorance = no prior information = unbiased



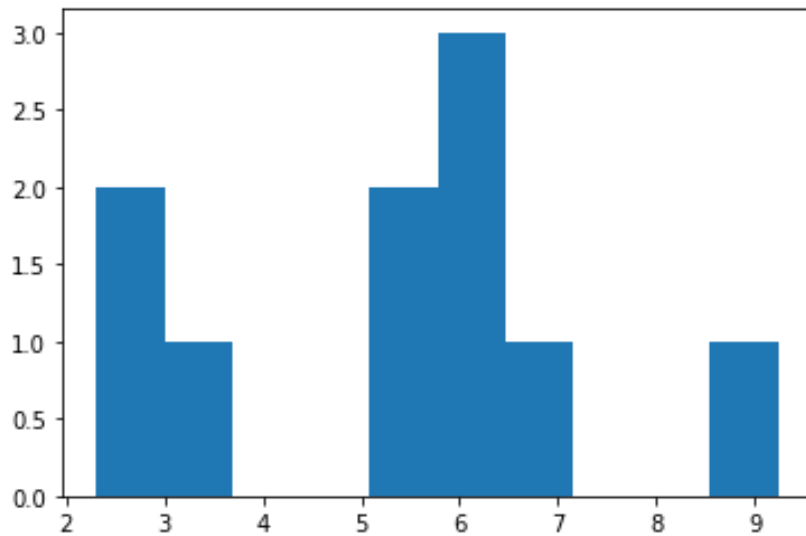
$$p(x) = \text{const}$$

plotting the pdf

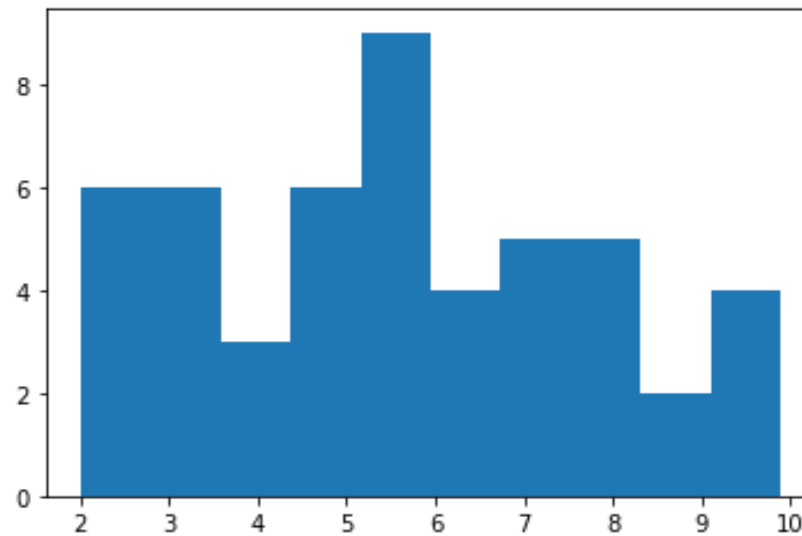
```
U = np.random.uniform(low, high, shape)  
plt.hist(U)
```

continuous support

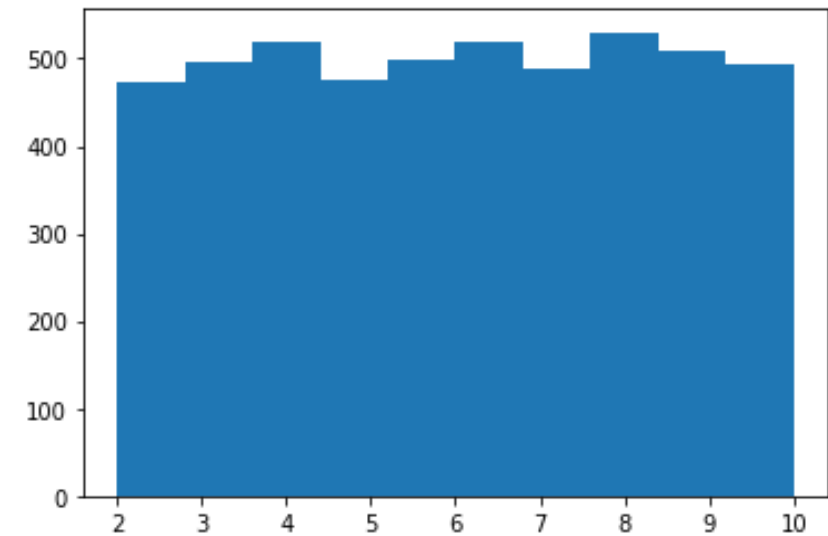
N = 10



N = 50



N = 5000





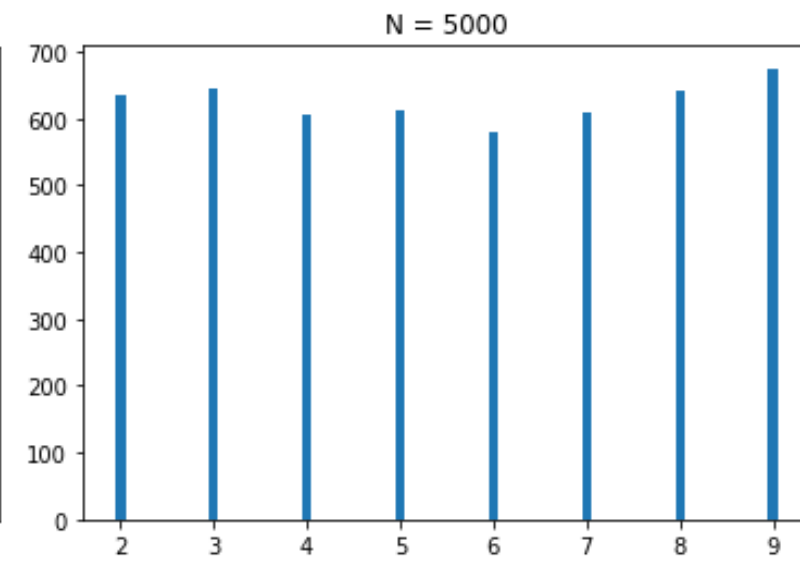
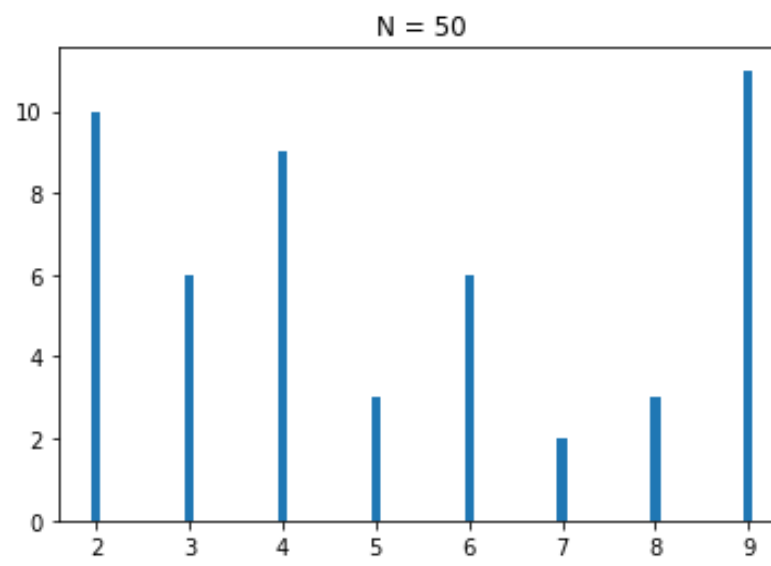
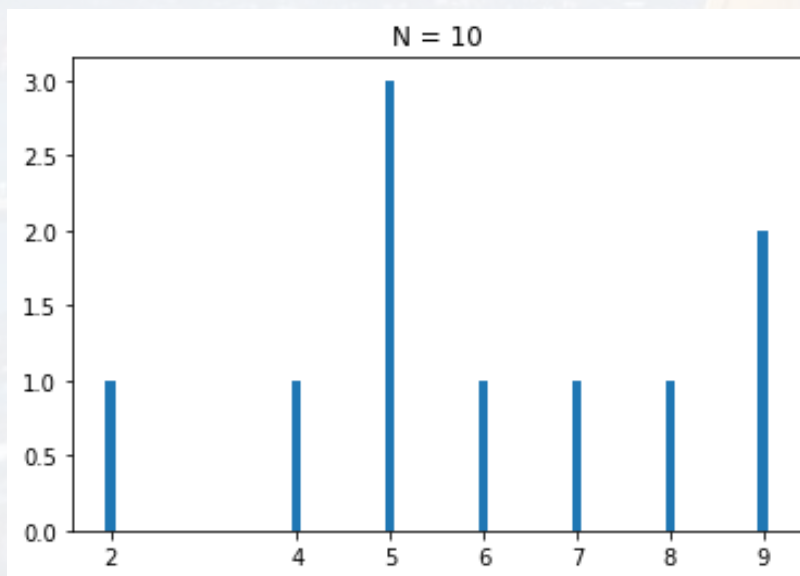
$$p(x) = \text{const}$$

plotting the pdf

```
U = np.random.randint(low, high, shape)
```

discrete support

```
labels, counts = np.unique(U, return_counts = True)
plt.bar(labels, counts, align = 'center', width = 0.1)
plt.gca().set_xticks(labels)
plt.title('N = ' + str(N))
```





$$p(x) = \text{const}$$

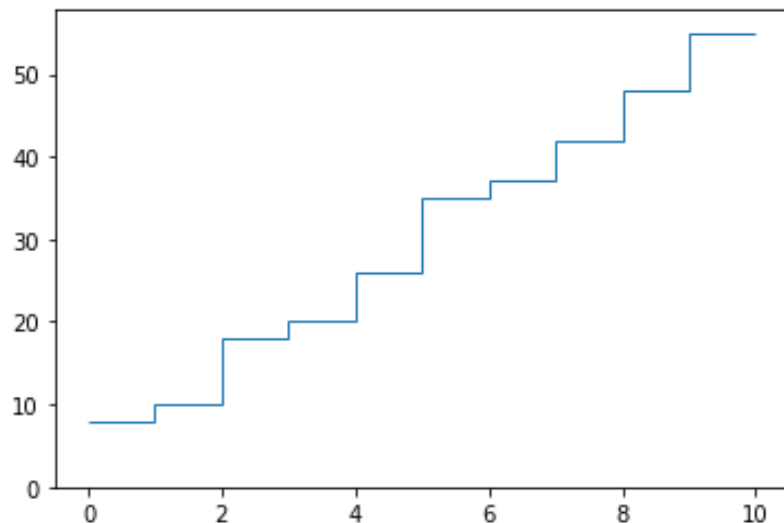
plotting the cdf

```
U = np.random.randint(low, high, shape)
```

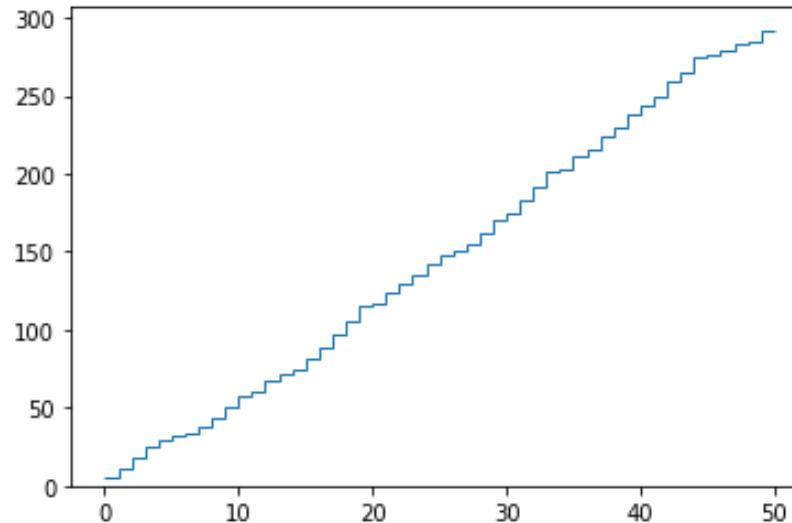
discrete support

```
C = np.cumsum(U)  
plt.stairs(C, baseline = None)  
plt.title('N = ' + str(N))
```

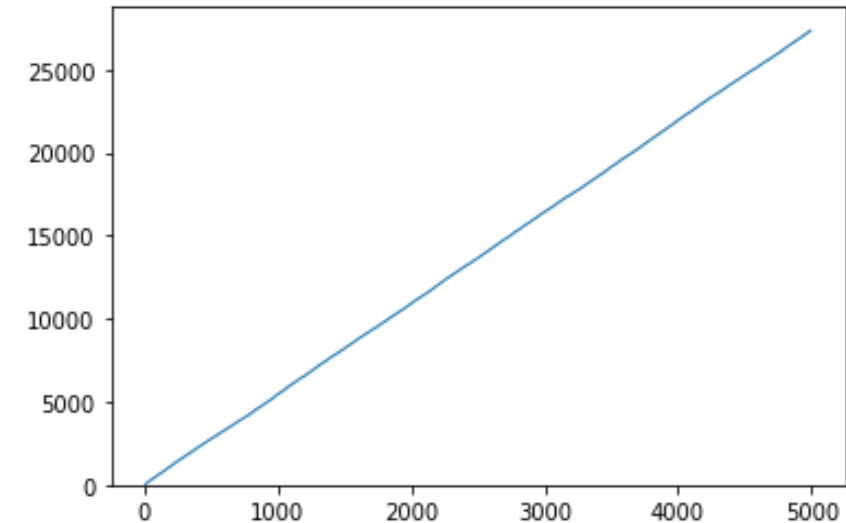
N = 10

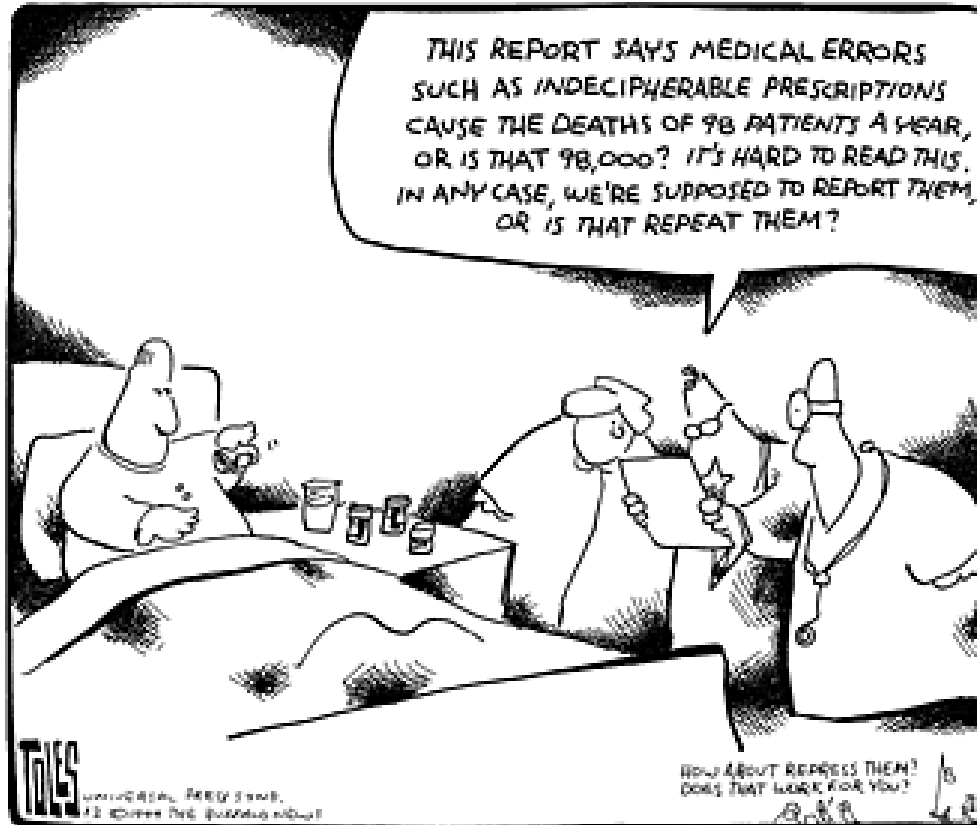


N = 50



N = 5000





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Outline:

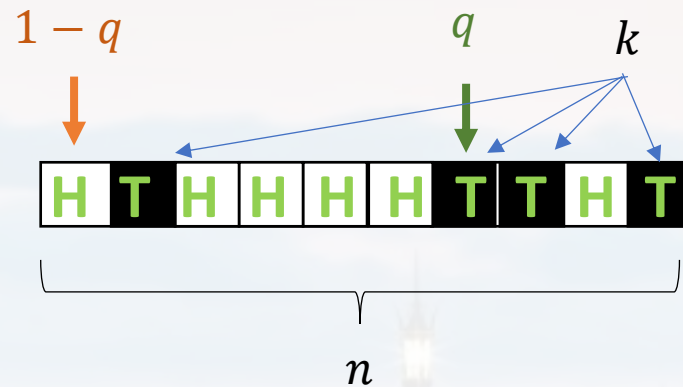
Basics

Most Common PDFs

- uniform
- **binomial**
- Poissonian
- Normal/Gaussian

Error Estimation

Bayesian Statistics



probability of having a sequence of **k tails** and **n-k heads**

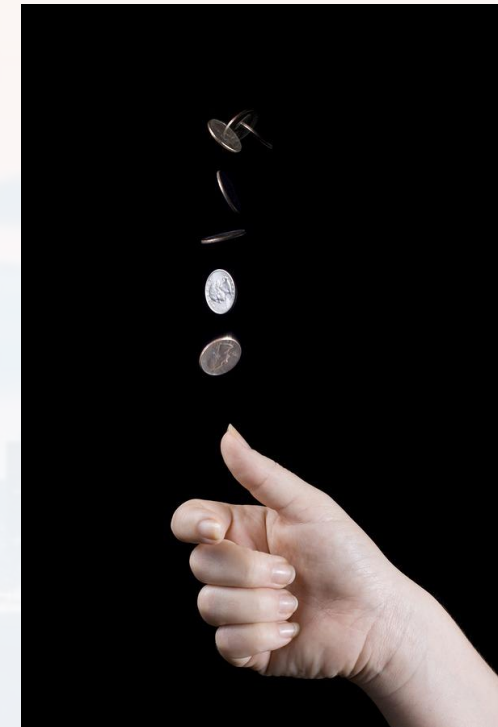
$$p_{tot} = \prod_i q_i^{n_i} = q^k (1 - q)^{n-k}$$

probability of having **any** sequence of **k tails** and **n-k heads**

$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution

$$\frac{n!}{k!(n-k)!} =: \binom{n}{k} \text{ ``n choose k''}$$



fair coin? $q = 0.5$???

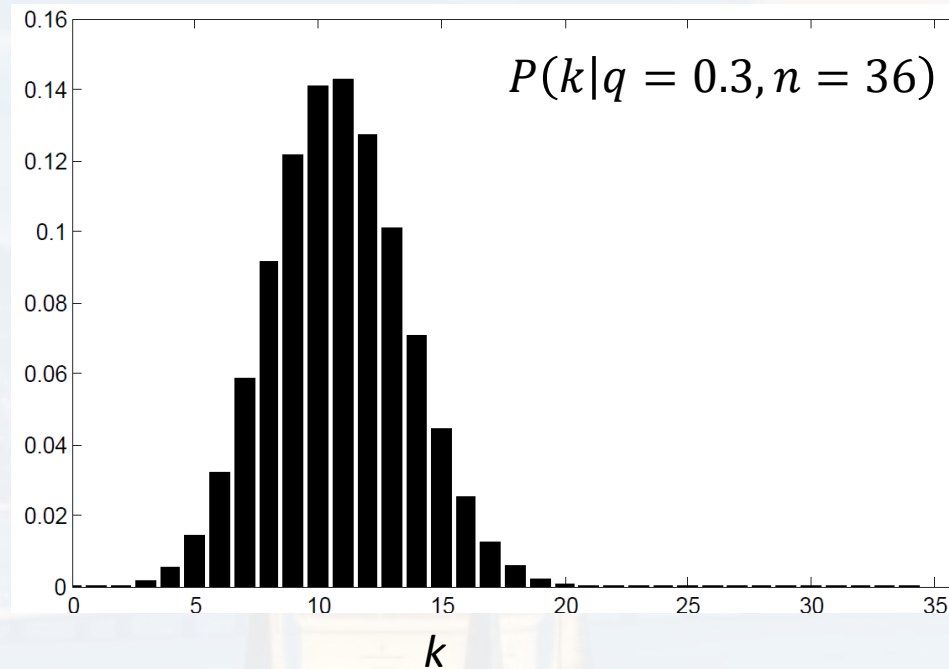


$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution

$$\mu = \sum_i x_i p(x_i)$$

$$\mu = \int x p(x) dx$$



$$\mu = \sum_{k=0}^n k \binom{n}{k} q^k (1 - q)^{n-k} = qn$$

$$\text{var}(k) = \sum_{k=0}^n (k - qn)^2 \binom{n}{k} q^k (1 - q)^{n-k} = qn(1 - q)$$



$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

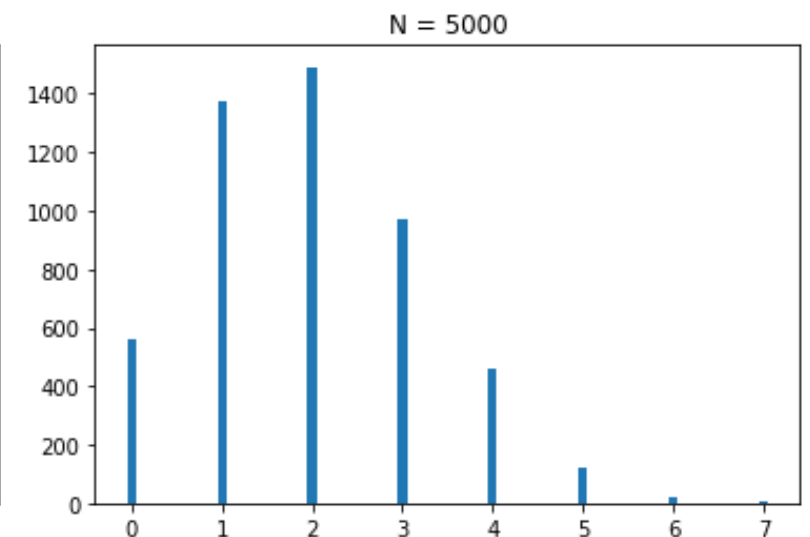
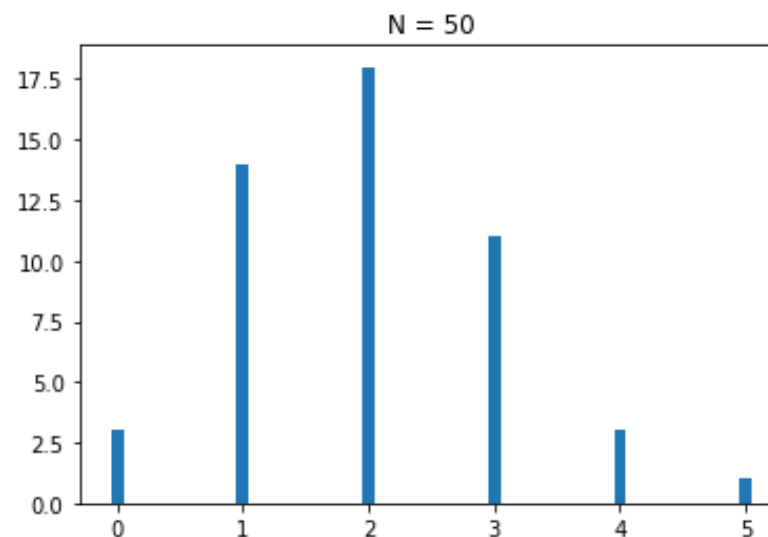
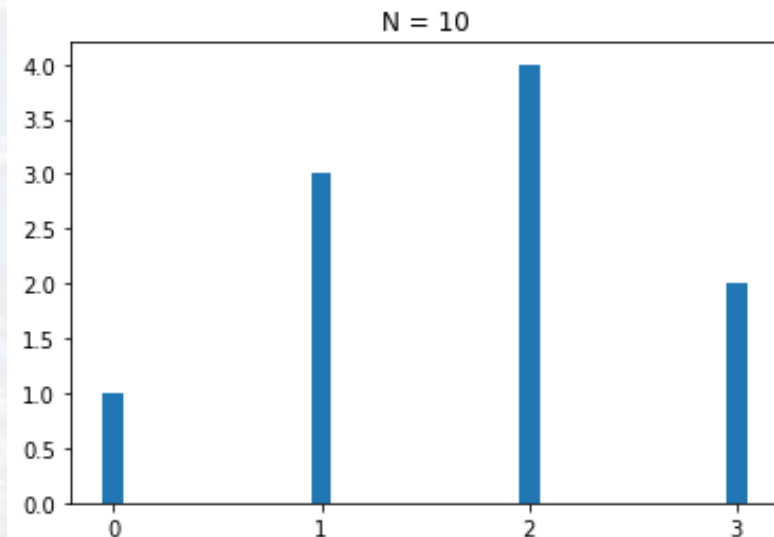
binomial distribution

$q = 0.2$

$n = 10$

```
K = np.random.binomial(n, q, N)
```

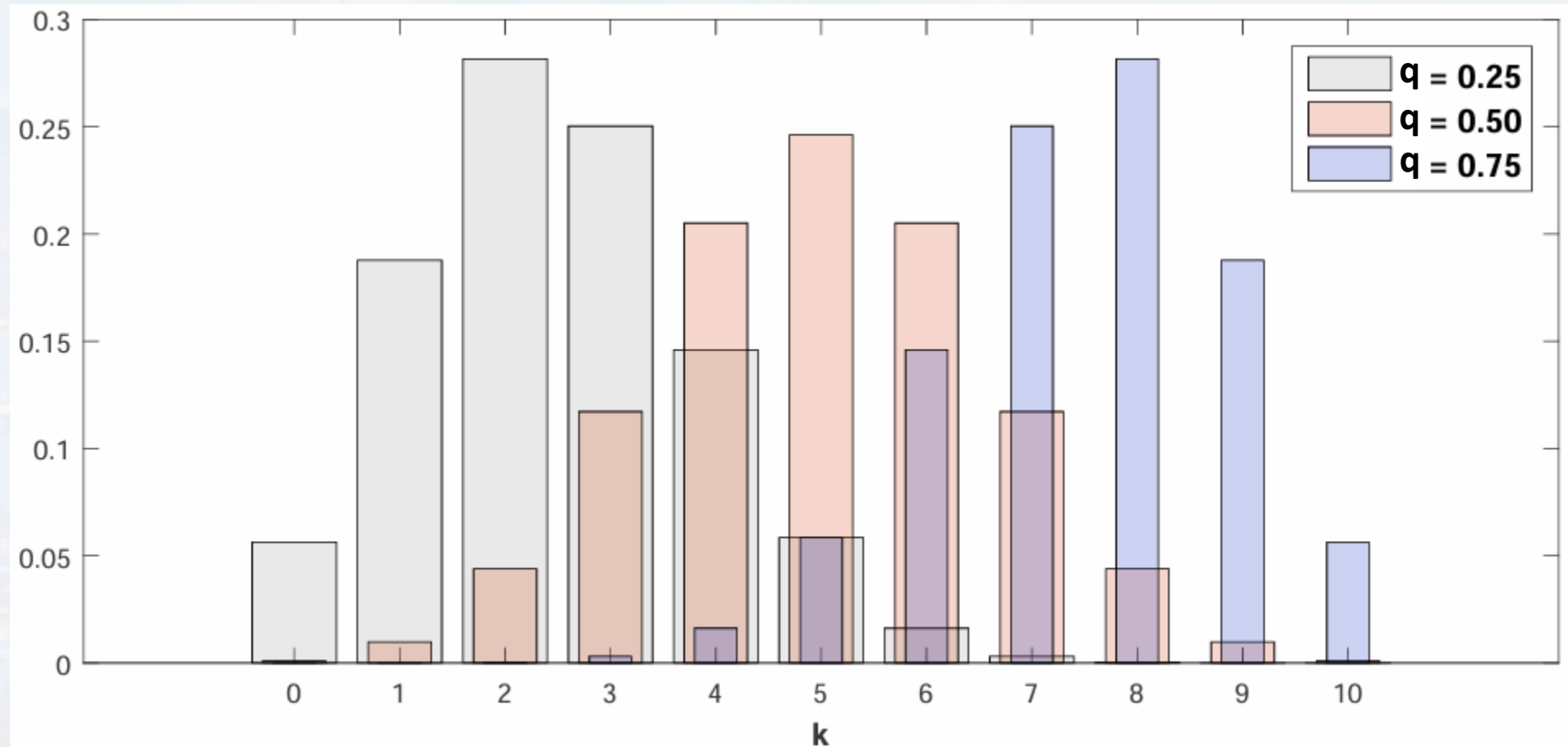
```
labels, counts = np.unique(K, return_counts = True)  
plt.bar(labels, counts, align = 'center', width = 0.1)  
plt.gca().set_xticks(labels)  
plt.title('N = ' + str(N))
```

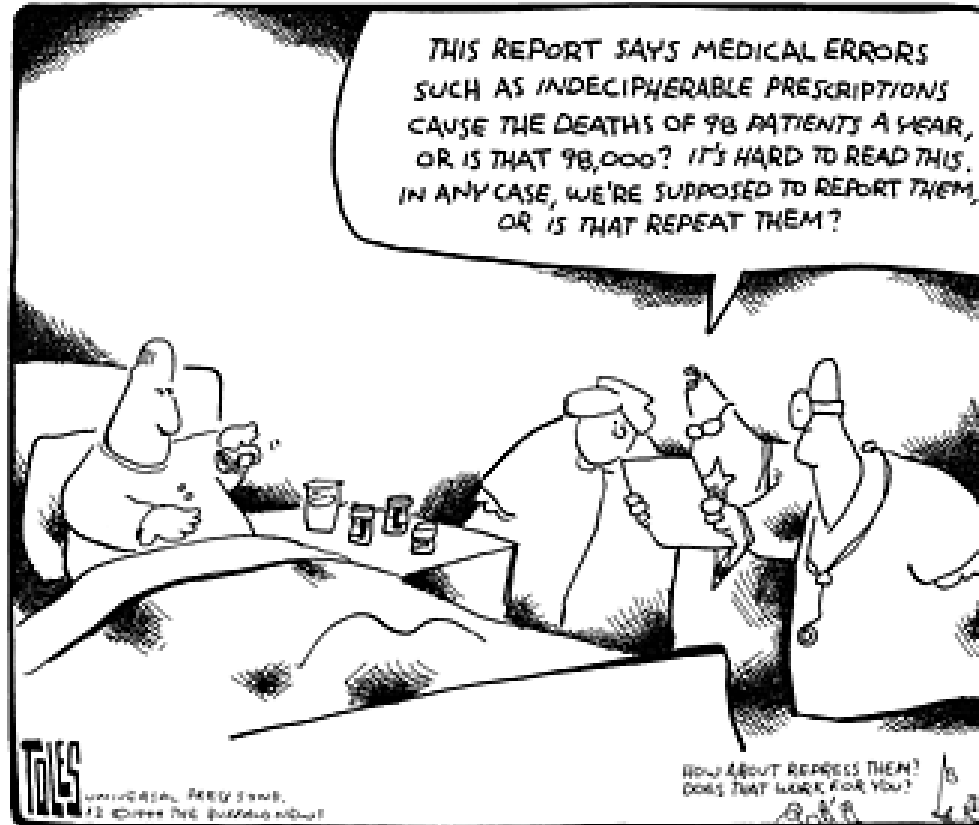




$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution





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Outline:

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$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution

rare events

$$\rightarrow q \ll 1$$

Taylor expansion for $(1 - q)^{n-k}$ around $q = 0$

$$(1 - q)^{n-k} = 1 - nq + \frac{(nq)^2}{2} - \frac{(nq)^3}{6} + \dots = e^{-nq}$$

$$\rightarrow n \rightarrow \infty$$

Stirling's approximation for $n!$

$$\frac{n!}{(n-k)!} \approx \sqrt{\frac{n}{n-k}} \frac{n^n e^{n-k}}{e^n (n-k)^{n-k}} \approx n^k$$

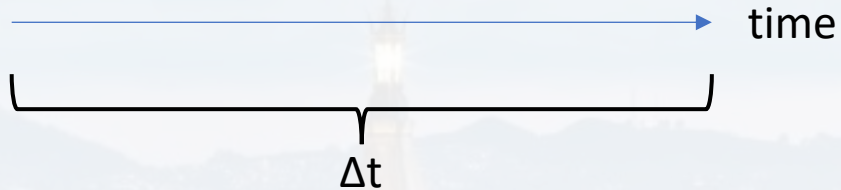
$$\binom{n}{k} q^k (1 - q)^{n-k} \approx \frac{(nq)^k e^{-nq}}{k!}$$



$$\binom{n}{k} q^k (1-q)^{n-k} \approx \frac{(nq)^k e^{-nq}}{k!}$$

often: $nq := \lambda$

events per time interval: $\lambda = c \Delta t$



rate $c = 4$ tails per Δt

$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

Poisson distribution

$$\mu = qn \rightarrow qn = \lambda$$

$$\text{var}(k) = qn(1-q) \rightarrow qn = \lambda$$





$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

Poisson distribution

$$\mu = \lambda$$

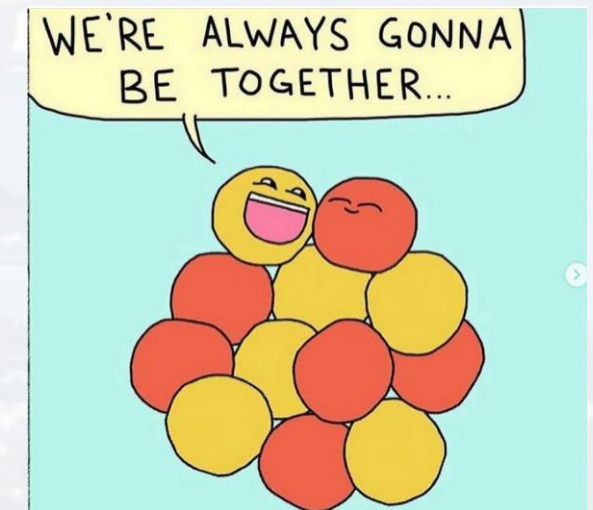
$$\text{var}(k) = \lambda$$

- **rare** events
- events are mutually **independent**
- events have **no duration**

examples:

- radioactive decay
- single photon detection
- lightning
- mutation of a gene
- receiving WhatsApp messages/SMS

rare: not that **a** atom decays,
→ that **this** atom decays within Δt





$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

Poisson distribution

$$\mu = \lambda$$

$$\text{var}(k) = \lambda$$

```
c      = 5  
delt = 10  
lam    = c * delt
```

```
K      = np.random.poisson(lam, N)
```

```
labels, counts = np.unique(K, return_counts = True)  
plt.bar(labels, counts, align = 'center', width = 0.1)  
plt.gca().set_xticks(labels)  
plt.title('N = ' + str(N))
```



$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

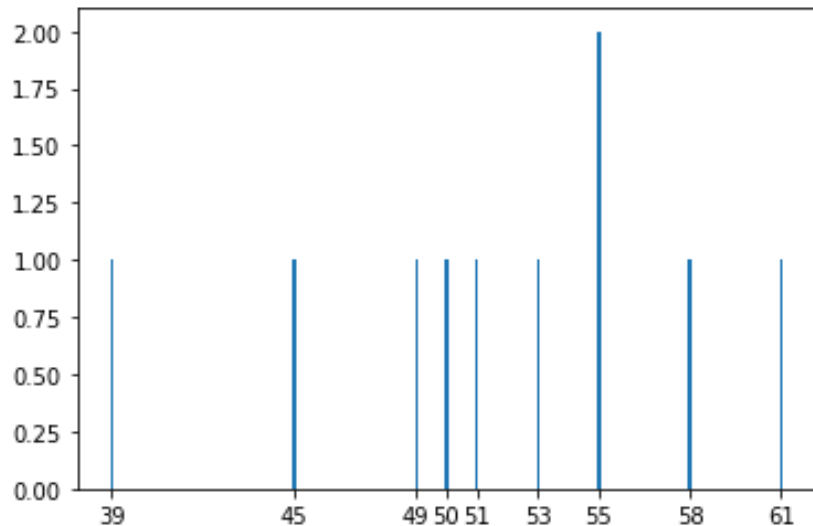
Poisson distribution

$$\mu = \lambda$$

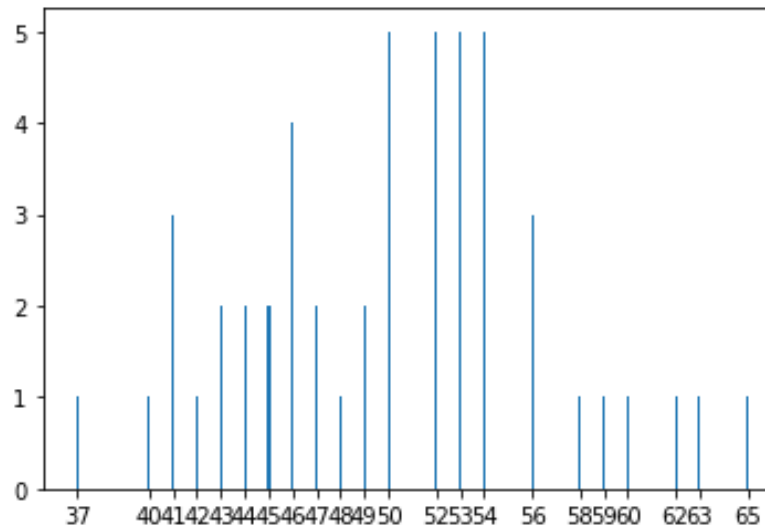
$$\text{var}(k) = \lambda$$

```
c = 5
delt = 10
lam = c * delt
K = np.random.poisson(lam, N)
labels, counts = np.unique(K, return_counts = True)
plt.bar(labels, counts, align = 'center', width = 0.1)
plt.gca().set_xticks(labels)
plt.title('N = ' + str(N))
```

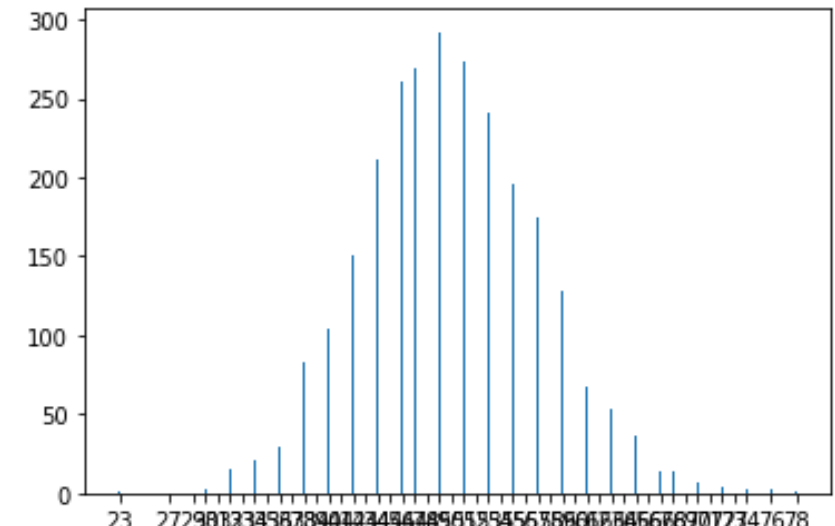
N = 10

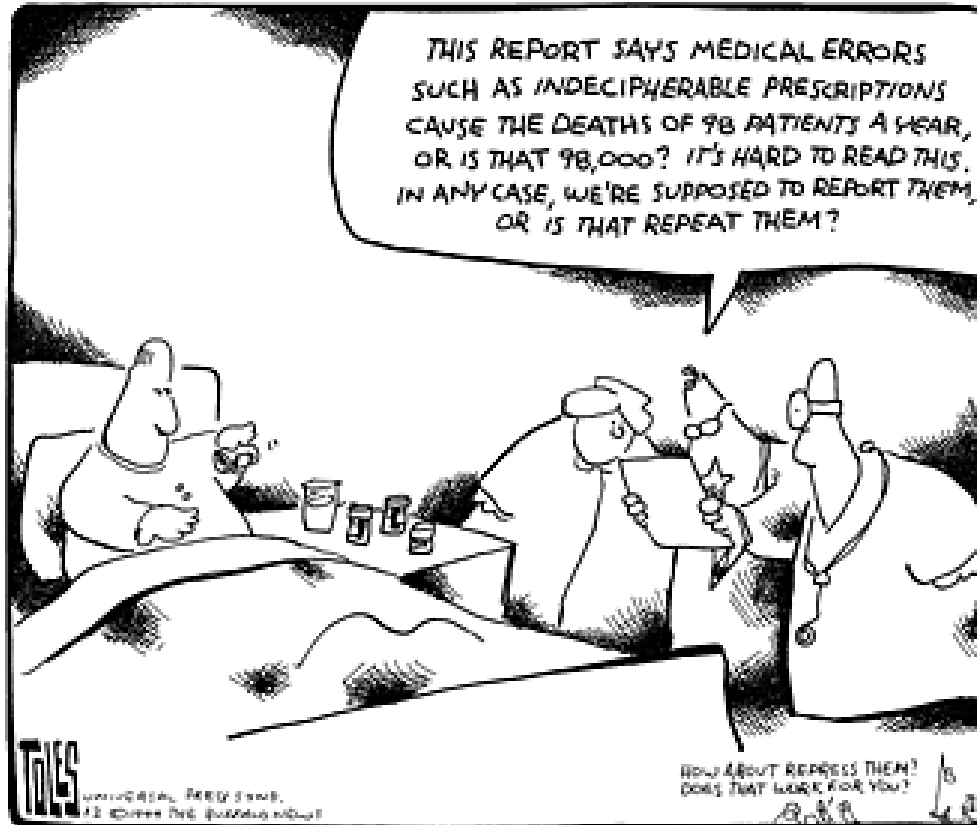


N = 50



N = 5000





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Outline:

Basics

Most Common PDFs

- uniform
- binomial
- Poissonian
- **Normal/Gaussian**

Error Estimation

Bayesian Statistics

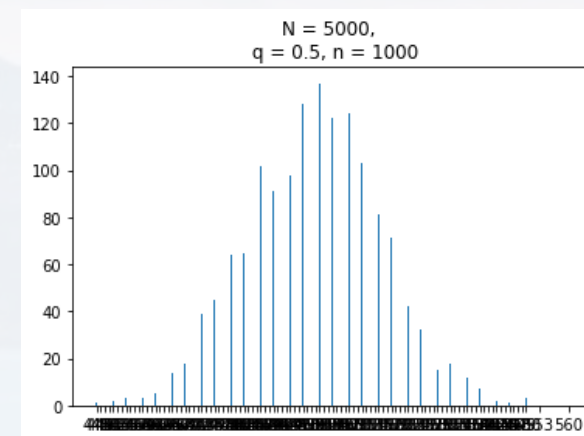
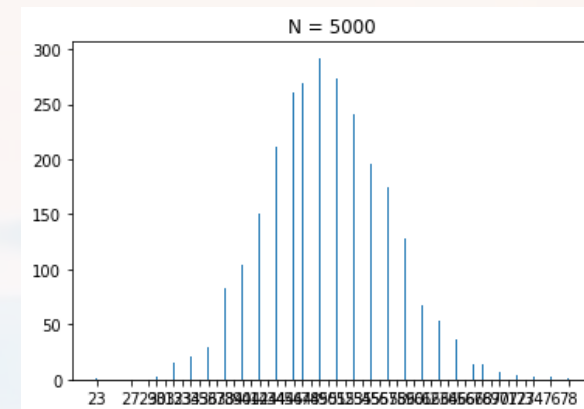


$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

Poisson distribution

$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

binomial distribution



Stirling's approximation for even larger n

$$P(k|n, p) \approx \frac{1}{\sqrt{2\pi nq(1-q)}} \exp\left[-\frac{(k - nq)^2}{2nq(1-q)}\right]$$



Stirling's approximation for even larger n

$$P(k|n, p) \approx \frac{1}{\sqrt{2\pi nq(1-q)}} \exp\left[-\frac{(k-nq)^2}{2nq(1-q)}\right]$$

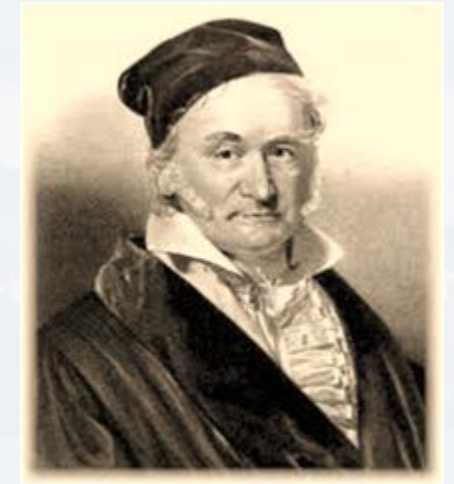
using $\sigma^2 = \text{var}(k) = qn(1-q)$

$$\mu = qn$$

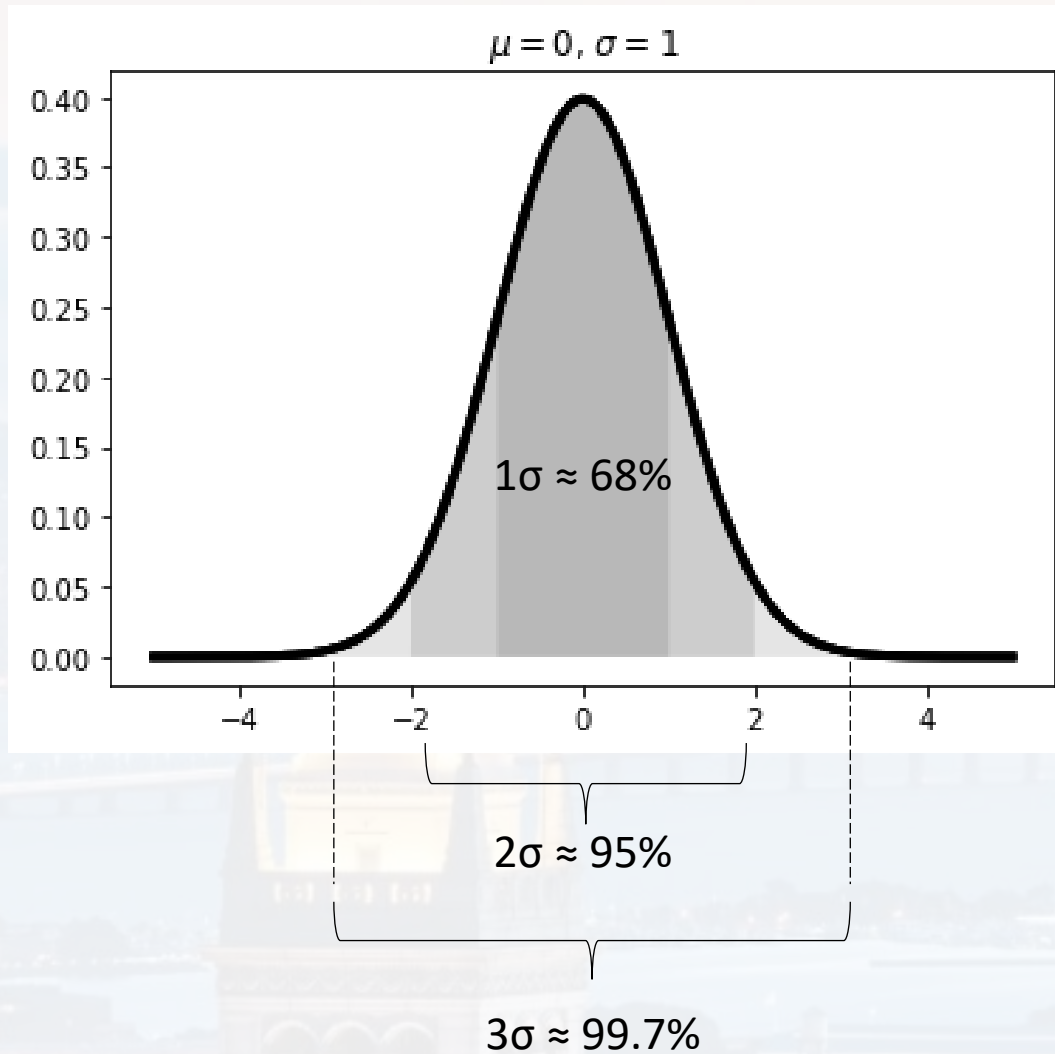
and $k := x$

$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi \sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2 \sigma^2}\right]$$

Normal/Gauss distribution

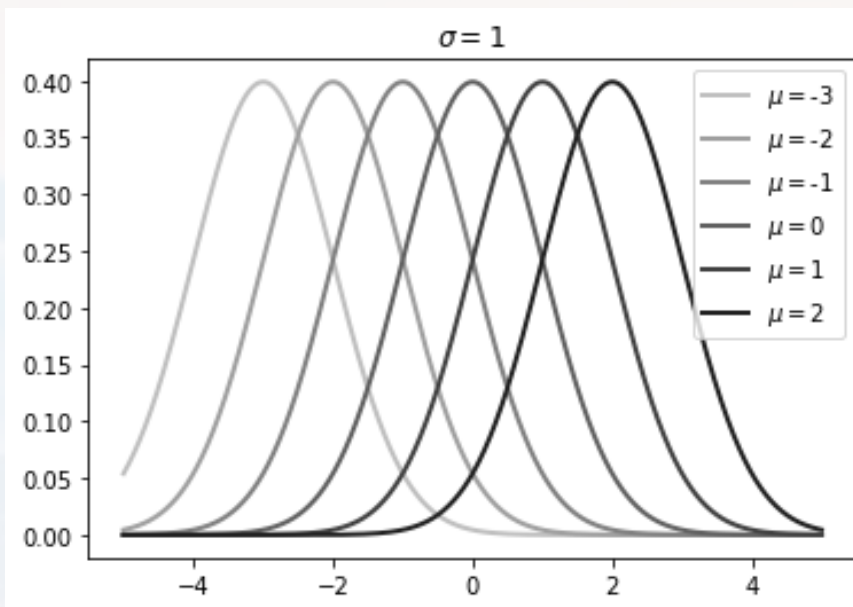


Note, that the **Poisson** and the **Binomial distribution** are *discrete*,
whereas the **Normal distribution** is *continuous*!



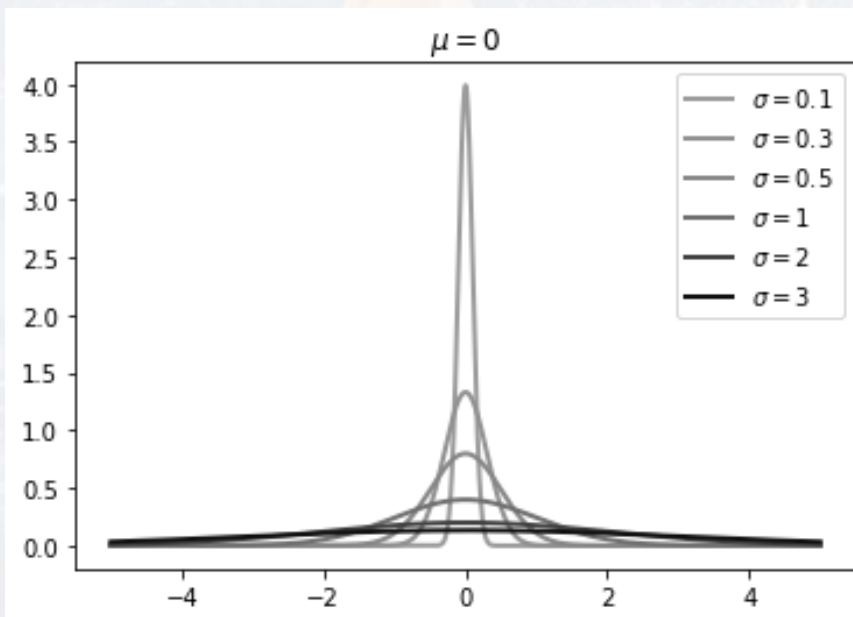
$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(x - \mu)^2}{2 \sigma^2} \right]$$

Normal/Gauss distribution



$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(x - \mu)^2}{2 \sigma^2} \right]$$

Normal/Gauss distribution

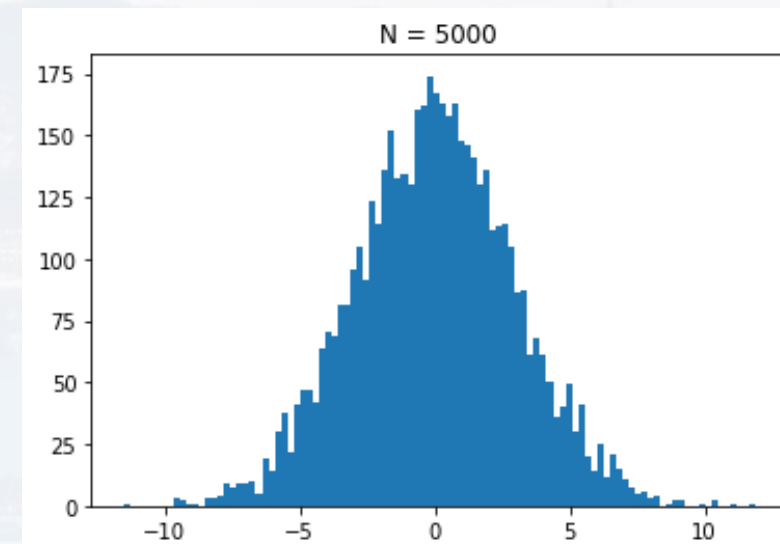
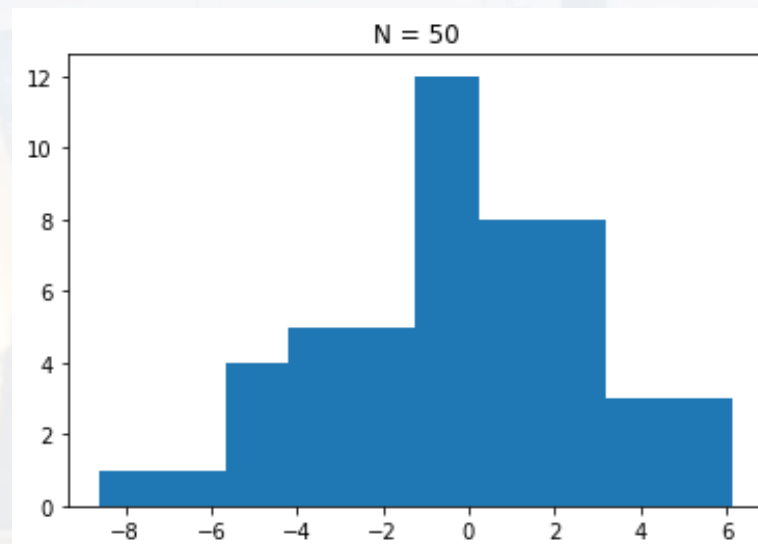
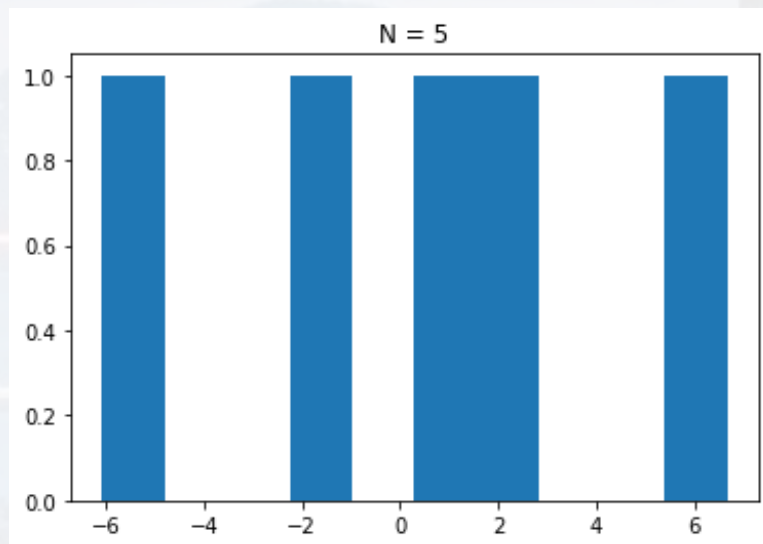


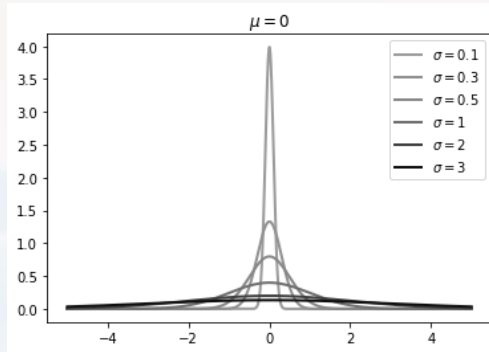


```
mu = 0  
s = 1  
P = np.random.normal(mu, s, N)  
plt.hist(P)  
plt.title('N = ' + str(N))
```

$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(x - \mu)^2}{2 \sigma^2} \right]$$

Normal/Gauss distribution





$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(x - \mu)^2}{2 \sigma^2} \right]$$

Normal/Gauss distribution

examples:

- diffusion processes
- approx. stat. error of data points
- approx. distribution of body height/shoe sizes/ weight, IQ
- approx. blood pressure, blood values
- approx. retirement age
-

applications:

- significance tests
- t-test
- ANOVA/MANOVA
- χ^2 - test
- χ^2 itself and students-t distribution
- ...

Why do so many quantities follow a normal distribution?



Why do so many quantities follow a normal distribution?

At the end... all probability distributions are **Maximum Entropy** Distributions, subject to a **set of constraints**

Distribution name	Probability density / mass function	Maximum Entropy constraint	Support
Uniform (discrete)	$f(k) = \frac{1}{b - a + 1}$	None	$\{a, a + 1, \dots, b - 1, b\}$
Uniform (continuous)	$f(x) = \frac{1}{b - a}$	None	$[a, b]$
Bernoulli	$f(k) = p^k (1 - p)^{1-k}$	$\mathbb{E}[K] = p$	$\{0, 1\}$
Geometric	$f(k) = (1 - p)^{k-1} p$	$\mathbb{E}[K] = \frac{1}{p}$	$\mathbb{N} \setminus \{0\} = \{1, 2, 3, \dots\}$
Exponential	$f(x) = \lambda \exp(-\lambda x)$	$\mathbb{E}[X] = \frac{1}{\lambda}$	$[0, \infty)$
Laplace	$f(x) = \frac{1}{2b} \exp\left(-\frac{ x - \mu }{b}\right)$	$\mathbb{E}[X - \mu] = b$	$(-\infty, \infty)$
Asymmetric Laplace	$f(x) = \frac{\lambda \exp\left(-(x - m) \lambda s \kappa^s\right)}{\left(\kappa + \frac{1}{\kappa}\right)}$ where $s \equiv \text{sgn}(x - m)$	$\mathbb{E}[(X - m) s \kappa^s] = \frac{1}{\lambda}$	$(-\infty, \infty)$
Pareto	$f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$	$\mathbb{E}[\ln X] = \frac{1}{\alpha} + \ln(x_m)$	$[x_m, \infty)$
Normal	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$	$\mathbb{E}[X] = \mu,$ $\mathbb{E}[X^2] = \sigma^2 + \mu^2$	$(-\infty, \infty)$



Why do so many quantities follow a normal distribution?

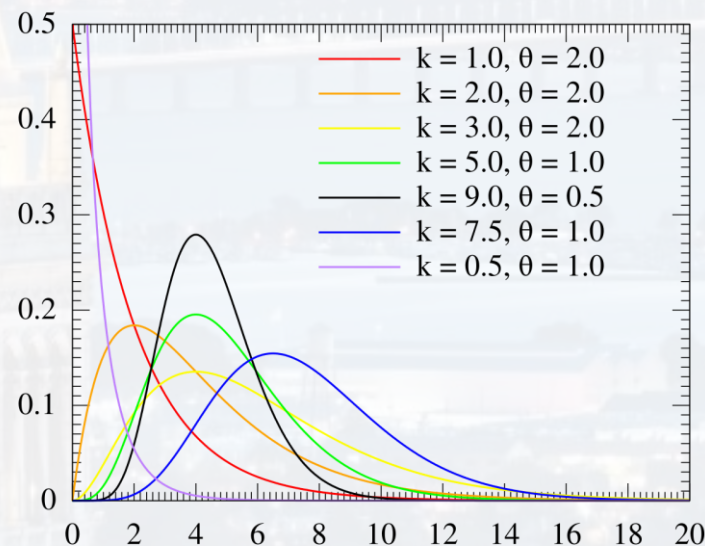
At the end... all probability distributions **are Maximum Entropy** Distributions, subject to a **set of constraints**

examples:

- approx. stat. error of data points
- approx. distribution of body height/shoe sizes/ weight, IQ
- approx. blood pressure, blood values
- approx. retirement age

....

Gamma	$f(x) = \frac{x^{k-1} \exp\left(-\frac{x}{\theta}\right)}{\theta^k \Gamma(k)}$	$\begin{aligned}\mathbb{E}[X] &= k\theta, \\ \mathbb{E}[\ln X] &= \psi(k) + \ln \theta\end{aligned}$	$[0, \infty)$
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binomial distribution

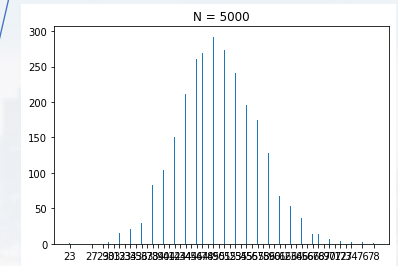
$$P(k|q, n) = \binom{n}{k} q^k (1 - q)^{n-k}$$

$q \rightarrow 0$

Poisson distribution

$$P(k|c) = \frac{(c \Delta t)^k e^{-c \Delta t}}{k!}$$

$n \rightarrow \infty$



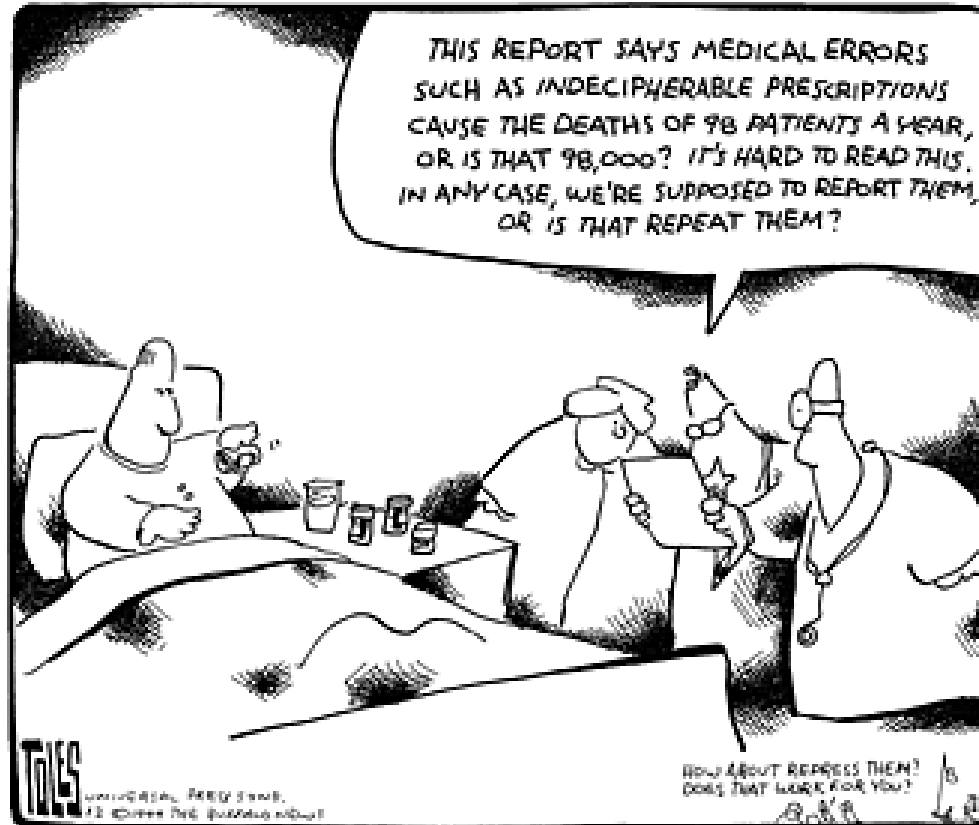
$n \rightarrow \infty$

Normal/Gauss distribution

$$P(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(x - \mu)^2}{2 \sigma^2} \right]$$

The fact that many datasets can be well approximated by a Normal distribution for $n \rightarrow \infty$ is called

Central Limit Theorem



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